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METHOD AND SYSTEM FOR DISPLAY OF AN ELECTRONIC REPRESENTATION OF PHYSICAL EFFECTS AND PROPERTY DAMAGE RESULTING FROM A PARAMETRIC NATURAL DISASTER EVENT

Abstract

A method for displaying an augmented reality (AR) representation of physical effects and property damage resulting from a parametric earthquake event is provided. The method includes scanning scenes with a common boundary and identifying the common boundary and objects in the first and second scenes. The method also includes creating an AR boundary for the common boundary and AR objects for the objects. The method further includes receiving a seismic characteristic. The method additionally includes, while scanning the first scene, displaying an aperture effect on the AR boundary and at least one seismic effect on the AR objects in the second scene based on the at least one received seismic characteristic, wherein the aperture effect allows viewing the seismic effect on AR objects in the second scene on the user device while the second scene is not visible to the one or more sensors of the user device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-in-part of U.S. patent application Ser. No. 18/513,213, filed Nov. 17, 2023, entitled METHOD AND SYSTEM FOR DISPLAY OF AN ELECTRONIC REPRESENTATION OF PHYSICAL EFFECTS AND PROPERTY DAMAGE RESULTING FROM A PARAMETRIC NATURAL DISASTER EVENT (Atty. Dkt. No. NPUI60-35809), now U.S. Pat. No. 12,229,909. Application Ser. No. 18/513,213 is a continuation of U.S. patent application Ser. No. 17/675,915, filed Feb. 18, 2022, entitled METHOD AND SYSTEM FOR DISPLAY OF AN ELECTRONIC REPRESENTATION OF PHYSICAL EFFECTS AND PROPERTY DAMAGE RESULTING FROM A PARAMETRIC NATURAL DISASTER EVENT (Atty. Dkt. No. NPUI60-35384), now U.S. Pat. No. 11,823,338. Application Ser. No. 17/675,915 claims benefit of U.S. Provisional Application No. 63/169,801, filed Apr. 1, 2021, entitled METHOD AND SYSTEM FOR DISPLAY OF AN ELECTRONIC REPRESENTATION OF PHYSICAL EFFECTS AND PROPERTY DAMAGE RESULTING FROM A PARAMETRIC NATURAL DISASTER EVENT (Atty. Dkt. No. NPUI60-35201) and U.S. Provisional Application No. 63/152,739, filed on Feb. 23, 2021, entitled METHOD AND SYSTEM FOR DISPLAY OF AN ELECTRONIC REPRESENTATION OF PHYSICAL EFFECTS AND PROPERTY DAMAGE RESULTING FROM A PARAMETRIC EARTHQUAKE EVENT (Atty. Dkt. No. NPUI60-34994). All of the aforementioned applications and their corresponding publications and their issued patents are hereby incorporated by reference herein in their entirety.

TECHNICAL FIELD
[0002] The present invention relates to techniques and technology for producing an electronic representation of the effects of a parametric earthquake event on a specific property comprising one or more structures, and more specifically to methods and systems of processing video and still images of the specific property taken during normal conditions to show the calculated movement and damage to the respective structures of the prope3004rty attributable to a parametric earthquake event.
BACKGROUND
[0003] The public is familiar with fictious depictions of earthquakes from motion pictures and even the depiction of actual earthquake events in documentary footage. However, such depictions do not serve to inform the public regarding the likely damage to their own property (both real and personal property) in the event of an earthquake event. Better information regarding the effects of earthquake events on property at a specific geographic location and the likely property damage resulting from such earthquake events can be useful in assessing the need for earthquake preparation.

[0004] A need therefore exists, for a system that can identify structures comprising property pictured at a specific geographic location and display an electronic representation, e.g., a video program and/or audiovisual program, showing a representation of the motion of the respective structures caused by a seismic event having known parameters, e.g., location, strength and/or duration.

[0005] A need further exists, for a system that can identify structures comprising a property pictured at a specific geographic location and display an electronic representation, e.g., still photo images, a video program and/or an audiovisual program showing a representation of damage to the respective structures caused by a seismic event having known parameters, e.g., location, strength and/or duration.

SUMMARY

[0006] In one aspect, a system receives images of property at a specified geographic location. The system processes, using an image recognition processor, the received images and identifies one or more structures comprising the property. The system retrieves parametric data regarding a designated seismic event. The system defines a key data pair corresponding to the specified geographic location and the designated seismic event. The system determines key attributes relating the key data pair. The system determines, using a computer or processor, values of a deemed seismic action at the specified geographic location by modifying the parametric data for the designated seismic event using the key attributes. The system produces images showing an electronic representation the one or more structures comprising the property moving in accordance with the deemed seismic action.

[0007] In one embodiment thereof, the images comprise a video program showing an electronic representation of the one or more structures moving in accordance with the deemed seismic action.

[0008] In another embodiment thereof, the images comprise an audiovisual program showing an electronic representation of the one or more structures moving in accordance with the deemed seismic action.

[0009] In yet another embodiment thereof, a key attribute determined by the system is the distance from the designated seismic event to the specified geographic location.

[0010] In still another embodiment thereof, a key attribute determined by the system is the intervening geology between the designated seismic event to the specified geographic location.

[0011] In a further embodiment thereof, the system assigns to one or more of the identified structures a respective structure attribute. The system determines values of deemed structure action for each respective structure by modifying the values of the deemed seismic action at the specified geographic location using the respective structure attribute. The system produces images showing an electronic representation of the one or more structures moving in accordance with the deemed seismic action and as further modified by the deemed structure action.

[0012] In a yet further embodiment thereof, the respective structure attribute is correlated to a determined mass of the structure.

[0013] In a still further embodiment thereof, the respective structure attribute is correlated to a determined resonant frequency of the structure.

[0014] In another aspect, a system receives images of property at a specified geographic location. The system processes, using an image recognition processor, the received images and identifies one or more structures comprising the property. The system receives parametric data regarding a designated seismic event. The system defines a key data pair corresponding to the specified geographic location and the designated seismic event. The system determines key attributes relating the key data pair. The system determines, using a computer or processor, values of a deemed seismic action at the specified geographic location by modifying the parametric data for the designated seismic event using the key attributes. The system determines a respective deemed damage corresponding to each structure based on the deemed seismic action. The system produces images showing an electronic representation the one or more structures comprising the property

modified with respective electronic representations of the respective deemed damage.
[0015] In one embodiment thereof, the images comprise a video program showing an electronic representation of the one or more structures modified to show the respective deemed damage.
[0016] In another embodiment thereof, the images comprise an audiovisual program showing an electronic representation of the one or more structures modified to show the respective deemed damage.
[0017] In yet embodiment thereof, the images comprise one or more still images showing an electronic representation of the one or more structures modified to show the respective deemed damage.
[0018] In still another embodiment thereof, a key attribute determined by the system is the distance from the designated seismic event to the specified geographic location.
[0019] In a further embodiment thereof, a key attribute determined by the system is the intervening geology between the designated seismic event to the specified geographic location.
[0020] In a yet further embodiment thereof, the system assigns to one or more of the identified structures a respective structure attribute. The system determines values of deemed structure damage for each respective structure by modifying the values of the deemed seismic action at the specified geographic location using the respective structure attribute. The system produces images showing an electronic representation of the one or more structures modified with the respective deemed damage corresponding to each structure based on the deemed seismic action and further corresponding to the respective structure attribute.
[0021] In a still further embodiment thereof, the respective structure attribute is correlated to a determined mass of the structure.
[0022] In another embodiment thereof, the respective structure attribute is correlated to a determined resonant frequency of the structure.
[0023] In yet another embodiment thereof, the respective structure attribute is correlated to a construction material of the structure.
[0024] In a third aspect, a method is provided for displaying an augmented reality (AR) representation of physical effects and property damage resulting from a parametric earthquake event. The method includes scanning, using one or more sensors of a user device, a scene in proximity to a user. The method also includes identifying a background and objects in the scene. The method further includes creating an AR background for the background of the scene and AR objects for the objects in the scene. The method further includes displaying the AR background and the AR objects on the display of the user device. In addition, the method includes receiving at least one seismic characteristic from the user through the display of the user device. The method additionally includes displaying at least one seismic effect on the AR objects and on the AR background in the scene displayed on the user device based on the at least one received seismic characteristic.
[0025] In one embodiment thereof, the background is a wall and the objects are picture frames.
[0026] In another embodiment thereof, displaying the at least one seismic effect on the AR objects comprises independently applying the at least one seismic effect on the AR objects and the AR background.
[0027] In another embodiment thereof, displaying the at least one seismic effect on the AR objects comprises independently applying the at least one seismic effect using a first effect type on the AR objects and using a second effect type on the AR background.
[0028] In yet another embodiment thereof, the first effect type is applying an AR oscillation and the second effect type is generating an AR crack.
[0029] In another embodiment thereof, independently applying the at least one seismic effect using a first effect type on the AR objects and using a second effect type on the AR background comprises selecting the first effect type based on a first identity of the AR objects and selecting the second effect type based on a second identity of the AR background.

[0030] In yet embodiment thereof, displaying the at least one seismic effect on the AR background comprises generating an AR crack on the AR background.
[0031] In yet another embodiment thereof, displaying the at least one seismic effect on the AR objects comprises generating an AR crack on at least one of the AR objects.
[0032] In yet another embodiment thereof, displaying the at least one seismic effect on the AR objects comprises applying an AR oscillation on at least one of the AR objects.
[0033] In still another embodiment thereof, displaying the at least one seismic effect on the AR objects comprises breaking an AR object into two or more AR partial objects, wherein each of the AR partial objects is smaller than the AR object.
[0034] In still another embodiment thereof, breaking the AR object into two or more AR partial objects comprises selecting respective sizes for the AR partial objects such that a cumulative size of the two or more partial objects is equal to a size of the AR object.
[0035] In a further embodiment thereof, displaying the at least one seismic effect on the AR objects comprises generating an AR crack on the AR background.
[0036] In a further embodiment thereof, generating an AR crack on the AR background comprises adding dynamic shading to the AR crack simulating parallax effects based on point of view.
[0037] In a yet further embodiment thereof, the method further includes displaying a control for a seismic intensity as one of the at least one seismic characteristic.
[0038] In a fourth aspect, a non-transitory computer readable medium is provided for displaying an augmented reality (AR) representation of physical effects and property damage resulting from a parametric earthquake event. The non-transitory computer readable medium contains instructions that when executed cause a processor to scan, using one or more sensors of a user device, a scene in proximity to a user. The non-transitory computer readable medium also contains instructions that when executed cause a processor to identify a background and objects in the scene. The non-transitory computer readable medium further contains instructions that when executed cause a processor to create an AR background for the background of the scene and AR objects for the objects in the scene. The non-transitory computer readable medium further contains instructions that when executed cause a processor to display the AR background and the AR objects on the display of the user device In addition, non-transitory computer readable medium contains instructions that when executed cause a processor to receive at least one seismic characteristic from the user through the display of the user device. The non-transitory computer readable medium additionally contains instructions that when executed cause a processor to display at least one seismic effect on the AR objects and on the AR background in the scene displayed on the user device based on the at least one received seismic characteristic.
[0039] In one embodiment thereof, the background is a wall and the objects are picture frames.
[0040] In another embodiment thereof, the instructions when executed cause the processor to display the at least one seismic effect on the AR objects comprise instructions that when executed cause the processor to independently apply the at least one seismic effect on the AR objects and the AR background.
[0041] In yet embodiment thereof, the instructions when executed cause the processor to display the at least one seismic effect on the AR objects comprise instructions that when executed cause the processor to generate a crack on at least one of the AR objects.
[0042] In still another embodiment thereof, the instructions when executed cause the processor to display the at least one seismic effect on the AR objects comprise instructions that when executed cause the processor to break an AR object into two or more AR partial objects. A cumulative size of the two or more partial objects is equal to a size of the AR object.
[0043] In a further embodiment thereof, the instructions when executed cause the processor to display the at least one seismic effect on the AR objects comprise instructions that when executed cause the processor to generate a crack on the AR background.
[0044] In a yet further embodiment thereof, the instructions when executed further cause the

processor to display a control for a seismic intensity as one of the at least one seismic characteristic.
[0045] In a fifth aspect, a user device is provided for displaying an augmented reality (AR) representation of physical effects and property damage resulting from a parametric earthquake event. The user device includes one or more sensors, a display, and a processor operably couple to the one or more sensors and the display. The one or more sensors are configured to scan a scene in proximity to a user. The display is configured to display the scene. The processor is configured to identify a background and objects in the scene. The processor is also configured to create an AR background for the background of the scene and AR objects for the objects in the scene. The processor is further configured to display the AR background and the AR objects on the display of the user device. The processor is further configured to receive at least one seismic characteristic from the user through the display of the user device. The processor is additionally configured to display at least one seismic effect on the AR objects and on the AR background in the scene displayed on the user device based on the at least one received seismic characteristic.
[0046] In one embodiment thereof, the background is a wall and the objects are picture frames.
[0047] In another embodiment thereof, the processor is further configured to independently apply the at least one seismic effect on the AR objects and the AR background.
[0048] In another embodiment thereof, the processor is further configured to apply the at least one seismic effect using a first effect type on the AR objects and using a second effect type on the AR background.
[0049] In yet another embodiment thereof, the first effect type is applying an AR oscillation and the second effect type is generating an AR crack.
[0050] In another embodiment thereof, independently applying the at least one seismic effect using a first effect type on the AR objects and using a second effect type on the AR background comprises selecting the first effect type based on a first identity of the AR objects and selecting the second effect type based on a second identity of the AR background.
[0051] In yet another embodiment thereof, displaying the at least one seismic effect on the AR background comprises generating an AR crack on the AR background.
[0052] In yet another embodiment thereof, to display the at least one seismic effect on the AR objects, the processor is further configured to generate an AR crack on at least one of the AR objects.
[0053] In still another embodiment thereof, to display the at least one seismic effect on the AR objects, the processor is further configured to break an AR object into two or more AR partial objects, wherein each of the AR partial objects is smaller than the AR object.
[0054] In still another embodiment thereof, to display breaking the AR object into two or more AR partial objects, the processor is configured to select respective sizes for the AR partial objects such that a cumulative size of the two or more partial objects is equal to a size of the AR object.
[0055] In a further embodiment thereof, to display the at least one seismic effect on the AR objects, the processor is further configured to generate an AR crack on the AR background.
[0056] In a further embodiment thereof, to generate an AR crack on the AR background, the processor is further configured to add dynamic shading to the AR crack simulating parallax effects based on point of view.
[0057] In a yet further embodiment thereof, the processor is further configured to display a control for a seismic intensity as one of the at least one seismic characteristic.
[0058] In a sixth aspect, a method is provided for displaying an augmented reality (AR) representation of physical effects and property damage resulting from a parametric earthquake event including a damage inventory and assessment. The method includes scanning, using one or more sensors of a user device, a scene in proximity to a user. The method also includes identifying a background and objects in the scene. The method further includes determining a number of the objects identified in the scene. The method further includes determining an area of the background

identified in the scene. The method further includes creating an AR background for the background of the scene and AR objects for the objects in the scene. The method further includes displaying the AR background and the AR objects on the display of the user device. In addition, the method includes receiving at least one seismic characteristic from the user through the user device. The method additionally includes displaying at least one seismic effect on the AR objects and on the AR background in the scene displayed on the user device based on the at least one received seismic characteristic. The method further includes determining a seismic damage factor based on the at least one received seismic characteristic. The method further includes determining an estimated object damage amount for the objects identified in the scene based on the determined number of objects and the determined seismic damage factor. The method further includes determining an estimated structure damage amount for the background identified in the scene based on the determined area of the background and the determined seismic damage factor.

[0059] In one embodiment, determining an estimated object damage amount further includes assigning a respective initial value to each respective object identified in the scene, determining a respective final value for each object based a respective assigned value and the respective determined seismic damage factor, and totaling the respective differences between the respective initial values and the respective final values for all the respective objects.

[0060] In another embodiment, the method further includes displaying the estimated object damage amount to the user.

[0061] In yet another embodiment, determining an estimated structure damage amount further includes assigning an initial unit value to the background identified in the scene, determining a respective final unit value for the background based an assigned unit value and the determined seismic damage factor, and multiplying the differences between the initial unit value and the final unit value by the determined area of the background.

[0062] In still another embodiment, the method further includes displaying the estimated structure damage amount to the user.

[0063] In another embodiment, the method further includes categorizing each identified object into one of at least two predetermined object classes, wherein a first initial value is assigned to each of the objects categorized in the first predetermined object class, and a second initial value is assigned to each of the objects categorized in the second predetermined object class.

[0064] In yet another embodiment, the method further includes displaying to the user a list of the respective objects identified in the scene and the respective initial values, allowing the user to provide a respective custom value for each respective object, and replacing the respective initial value with the respective custom value as the respective assigned value used to determine the respective final value of the respective object.

[0065] In still another embodiment, the method further includes displaying to the user the initial unit value for the background, allowing the user to provide a custom unit value for the background, and replacing the initial unit value with the custom unit value as the assigned unit value used to determine the final unit value for the estimated structure damage.

[0066] In another embodiment, the method further includes creating a first list of objects identified in a first scene and their respective assigned values, creating a second list of objects identified in a second scene and their respective assigned values, and combining the first list and the second list into a consolidated object list including all the object identified in the first scene and in the second scene and their respective assigned values.

[0067] In yet another embodiment, the method further includes using the consolidated object list to determine a consolidated object damage value for all the objects identified in the first scene and in the second scene and displaying the consolidated object damage value to the user.

[0068] In a seventh embodiment, a method for displaying an augmented reality (AR) representation of physical effects and property damage resulting from a parametric earthquake event is provided. The method includes scanning, using one or more sensors of a user device, first and second scenes

with a common boundary. The method also includes identifying the common boundary and objects in the first and second scenes. The method further includes creating an AR boundary for the common boundary of the first and second scenes and AR objects for the objects in the first and second scenes. The method additionally includes receiving at least one seismic characteristic from a user of the user device. The method also includes, while scanning the first scene with the user device, displaying an aperture effect on the AR boundary and at least one seismic effect on the AR objects in the second scene based on the at least one received seismic characteristic, wherein the aperture effect allows viewing the at least one seismic effect on AR objects in the second scene on the user device while the second scene is not visible to the one or more sensors of the user device.
[0069] In an eighth embodiment, non-transitory computer readable medium, for displaying an augmented reality (AR) representation of physical effects and property damage resulting from a parametric earthquake event, containing instructions that when executed cause a processor to scan, using one or more sensors of a user device, first and second scenes with a common boundary. The instructions that when executed also cause the processor to identify the common boundary and objects in the first and second scenes. The instructions that when executed further cause the processor to create an AR boundary for the common boundary of the first and second scenes and AR objects for the objects in the first and second scenes. The instructions that when executed also cause the processor to receive at least one seismic characteristic from a user of the user device. The instructions that when executed also cause the processor to while scanning the first scene with the user device, display an aperture effect on the AR boundary and at least one seismic effect on the AR objects in the second scene based on the at least one received seismic characteristic, wherein the aperture effect allows viewing the at least one seismic effect on AR objects in the second scene on the user device while the second scene is not visible to the one or more sensors of the user device
[0070] In a ninth embodiment, a user device includes one or more sensors and a processor. The one or more sensors are configured to scan first and second scenes with a common boundary. The processor is configured to identify the common boundary and objects in the first and second scenes. The processor is also configured to create an AR boundary for the common boundary of the first and second scenes and AR objects for the objects in the first and second scenes. The processor is further configured to receive at least one seismic characteristic from a user of the user device. The processor is additionally configured to while scanning the first scene with the user device, display an aperture effect on the AR boundary and at least one seismic effect on the AR objects in the second scene based on the at least one received seismic characteristic, wherein the aperture effect allows viewing the at least one seismic effect on AR objects in the second scene on the user device while the second scene is not visible to the one or more sensors of the user device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0071] For a more complete understanding, reference is now made to the following description taken in conjunction with the accompanying Drawings in which:
[0072] FIG. **1** is a block diagram of a system for display of an electronic representation of physical effects and property damage resulting from a parametric earthquake event in accordance with this disclosure;
[0073] FIG. **2** is a map showing the position of a parametric earthquake event and several geographic locations;
[0074] FIG. **3** shows an image of an actual property comprising multiple structures;
[0075] FIG. **4** shows an image of an electronic representation of the multiple structures of the property of FIG. **3** moving in accordance with the deemed seismic action from a parametric

earthquake event;
[0076] FIG. **5** shows an image of an electronic representation of the multiple structures of the property of FIG. **3** modified with electronic representations of deemed damage from a parametric earthquake event;
[0077] FIG. **6** shows an image of actual roofing of a structure;
[0078] FIG. **7** shows a first damage texture representing damaged roofing;
[0079] FIG. **8**A shows a first texture map corresponding to a first amount of deemed damage, and FIG. **8**B shows the first texture map applied to the first damage texture to produce an electronic representation of a first level of deemed damage;
[0080] FIG. **9**A shows a second texture map corresponding to a second amount of deemed damage, and FIG. **9**B shows the second texture map applied to the first damage texture to produce an electronic representation of a second level of deemed damage;
[0081] FIG. **10**A shows the image of the structure roofing of FIG. **6** modified using the electronic representation of the first level of deemed damage to produce an electronic representation of the structure damaged in accordance with the first level of deemed damage;
[0082] FIG. **10**B shows the image of the structure roofing of FIG. **6** modified using the electronic representation of the second first level of deemed damage to produce an electronic representation of the structure damaged in accordance with the second level of deemed damage;
[0083] FIG. **11** shows an image of an actual structure on a property;
[0084] FIG. **12** shows a wireframe representation of the structure of FIG. **11** after recognition of the original image by the image recognition processor and processing into a wireframe by the graphics modeling processor;
[0085] FIG. **13** shows a damaged wireframe representation produced by modifying the original wireframe representation of the structure of FIG. **11** in accordance with an assigned level of deemed damage received from the system processor;
[0086] FIG. **14** shows an electronic representation of the structure of FIG. **11** damaged in accordance with the deemed damage level by rendering texture on the damaged wireframe representation and modifying the original texture with an electronic representation of damage in the texture in accordance with the assigned level of deemed damaged;
[0087] FIGS. **15**A-**15**C are block diagrams showing a method for implementing, on a user device, a system for display of an electronic representation of physical effects and property damage resulting from a parametric earthquake event in accordance with this disclosure;
[0088] FIGS. **16**A-**16**E are block diagrams showing a method for implementing, on a system server, a system for display of an electronic representation of physical effects and property damage resulting from a parametric earthquake event in accordance with this disclosure;
[0089] FIG. **17** is an image of an interior room of a structure in accordance with this disclosure;
[0090] FIGS. **18**A-**18**E are user interfaces for display of an electronic representation of physical effects and property damage resulting from a parametric natural disaster event in accordance with this disclosure;
[0091] FIG. **19** is a block diagram of a system for display of an electronic representation of physical effects and property damage resulting from a parametric earthquake event in accordance with this disclosure;
[0092] FIGS. **20**A and **20**B illustrate an example method for an AR earthquake visualization and assessment process in accordance with this disclosure;
[0093] FIGS. **21**A and **21**B illustrate an example method for a scene generation process in accordance with this disclosure;
[0094] FIG. **22** illustrates an example meshes for assigning to a plane in accordance with this disclosure;
[0095] FIGS. **23**A and **23**B illustrate an example object extraction from mesh in accordance with this disclosure;

[0096] FIG. **24** illustrates an example method for a crack creation algorithm in accordance with this disclosure;
[0097] FIGS. **25**A and **25**B illustrate an example UV mapping for the crack creation algorithm in accordance with this disclosure;
[0098] FIGS. **26**A-**26**C illustrate an example Voronoi noise for the crack creation algorithm in accordance with this disclosure;
[0099] FIG. **27** illustrates an example AR earthquake visualization and assessment system in accordance with this disclosure, namely, from the viewpoint of a user holding a mobile device (the user's hand is not shown) displaying an AR earthquake scene based on the actual room scene visible behind the mobile device;
[0100] FIGS. **28**A and **28**B illustrate example inventory tables for objects and backgrounds identified in the scene in accordance with this disclosure;
[0101] FIG. **29** illustrates an example method for an AR earthquake visualization and assessment process in accordance with this disclosure;
[0102] FIGS. **30**A-**30**C illustrate an example AR earthquake visualization and assessment system in accordance with this disclosure, namely, from the viewpoint of a user holding a mobile device (the user's hand is not shown) displaying an AR earthquake s30Acene based on the actual room scene visible behind the mobile device;
[0103] FIG. **31** illustrates an example method for an AR earthquake visualization and assessment process in accordance with this disclosure.

DETAILED DESCRIPTION

[0104] Referring now to the drawings, wherein like reference numbers are used herein to designate like elements throughout, the various views and embodiments of method and systems for display of an electronic representation of physical effects and property damage resulting from a parametric natural disaster event are illustrated and described, and other possible embodiments are described. The figures are not necessarily drawn to scale, and in some instances the drawings have been exaggerated and/or simplified in places for illustrative purposes only. One of ordinary skill in the art will appreciate the many possible applications and variations based on the following examples of possible embodiments.
[0105] Referring to FIG. **1**, there is illustrated a system **100** for display of an electronic representation of physical effects and property damage resulting from a parametric earthquake event in accordance with one aspect. Unless otherwise noted, in this application the terms “parametric earthquake event” and “parametric seismic event” have the same meaning. A parametric earthquake event is a set of parameters and values for defining an earthquake event in terms of scientific principles, e.g., physics and geological principles. Such parameters can include, but are not limited to, date, time, epicenter location, hypocenter location (i.e., focus), duration, magnitude, peak ground acceleration (“PGA”), and maximum shaking intensity (“SI”), maximum ground amplitude, mean ground amplitude, shaking frequency, soil type, rock type and fault classification. A parametric earthquake event can include some or all parameter values measured during an actual, i.e., “historical” earthquake event, but can also include some or all parameters values selected by a user. The system **100** includes a system server **102** or computing device. The system server **102** contains one or more processors, RAM memory, storage units and communication interfaces for sending and receiving data to and from other devices. The system server **102** may comprise a single machine or multiple machines, including virtual machines resident on “cloud servers.” The system server **102** is operably connected to a communication network **104** and to one or more subsystems, e.g., an image recognition processor **106**, a graphics animation processor **108** and/or an image rendering engine **110**. In some embodiments, the communication network **104** is the internet; however, the communication network **104** can be any type of network that allows the system server **102** and subsystems **106**, **108** and **110** to communicate with one another and with a user device **114**. In the embodiment illustrated in FIG. **1**,

the server **102** is directly connected to the subsystems **106**, **108** and **110**; however, in other embodiments some or all of the subsystems may be connected to the server via the communication network **104**. In some embodiments, the subsystems **106**, **108** and **110** can communicate directly with one another for improved data transfer.

[0106] The image recognition processor **106** can include or incorporate an image labeling or annotation tool (i.e., "labeling tool"). The labeling tool can be used to process and label the images for bounding box object detection and segmentation so that the image is readable by machines. In some embodiments, the labeling tool can utilize human assistance and in other embodiments the labeling tool can operate solely with machine learning or artificial intelligence processes. In some embodiments, different image labeling tools may be used for processing images of different image types (e.g., interior images, exterior images, etc.). Using the image labeling tools, the various objects in the provided image (e.g., user image **300**, exterior image **1100**, interior image **1700**) can be labeled for specific purposes. In some embodiments, labeled objects can be selected for replacement by a computer graphic object (e.g., 2D sprite or 3D polygon), which can be moved on-screen and otherwise manipulated as a single entity, e.g., for purposes of event animation. In some embodiments, labeled objects may be classified into different types or categories of objects for different purposes. For example, in some embodiments, labeled objects can be categorized for properties relating to event animation, e.g., movable-type objects, bendable-type objects, breakable-type objects, waterproof-type objects, water damageable-type objects, etc. In other embodiments, labeled objects can be categorized for properties relating to inventory or damage assessment, e.g., table-type objects, chair-type objects, window-type objects, hanging art-type objects, TV/computer screen-type objects. Such classification may be performed by the image recognition processor **106** or by another processor, e.g., the system processor **102** or graphics animation processor **108**. The image labeling tools can use known or future-developed detection techniques for detection of the object including, but not limited to, semantic, bounding box, key-point and cuboid techniques.

[0107] A seismic event database **112** is operably connected to the server **102**, either via the communication network **104** or directly. The seismic event database **112** stores parametric seismic event data corresponding to one or more earthquakes or other seismic events. The parametric seismic event data can include, but is not limited to, values for the following parameters: event name, event date, event epicenter location, event focus (i.e., hypocenter) location, event duration, event magnitude, event PGA, event maximum shaking intensity ("SI"), event maximum ground amplitude, event mean ground amplitude, and event shaking frequency for each seismic event. The seismic events data in the seismic event database **112** may be actual historical earthquake data, "relocated" earthquake data (i.e., where the majority of the data corresponds to a historical earthquake, but the epicenter/hypocenter location is changed to a different location) or hypothetical earthquake data specified by a user or otherwise generated. In some embodiments, the seismic event database **112** can be located within the server **102** or one of the subsystems. In some embodiments, the seismic event database **112** can include data and/or values from public or private earthquake reporting agencies, such as the U.S. Geological Survey.

[0108] User devices **114**, **115** can connect to the system **100** through the communication network **104**. The user devices **114**, **115** can be mobile devices such as mobile phones, tablets, laptop computers or they can be stationary devices such as desktop computers or smart appliances including, but not limited to, smart televisions. In some embodiments, aspects of the current system **100** may include downloadable software or "apps" resident on the user devices **114**, **115** and/or non-downloadable software (e.g., "cloud based software") that remains resident on the server **102** or other elements of the system **100** and is accessed by the user device **114** via a web browser or other network interface.

[0109] Using the user devices **114**, **115**, system users can upload images of actual property via the network **104** to the server **102**. The images of actual property can be captured using a camera **116** on the first user device **114** or from images stored in memory **118**, e.g., a computer memory, hard drive, flash drive or other data storage technology. The images can be video programs, audiovisual programs and/or single or multiple still photo images. User devices **114** can also transmit additional information to the server **102** regarding the images, including, but not limited to, the geographic location of the property in the images (either entered by the user as text, captured via GPS or wireless location information on the user device **114**, or captured vis geotagging information on the image file), the address of the property in the images, the name and/or other contact information of the user, a desired parametric seismic event and/or desired seismic parameters to be used in creating a parametric seismic event. Desired seismic parameters can include strength of the seismic event, duration of the seismic event, etc. Each of the seismic parameters can be input on the user devices **114**, **115** by any suitable means. For example, the seismic parameters can be input or selected using a predetermined list, a slider associated with different values of the seismic parameters, a knob, a number entry. The inputs can be physical components on the user device **114**, **115** or virtual representations on a display **120** of the user device **114**, **115**.

[0110] The system **100** processes, using an image recognition processor **106**, the received images and identifies one or more structures comprising the property. The system **100** retrieves parametric data regarding a designated seismic event from the seismic event database **112**. The system **100** defines a key data pair corresponding to the specified geographic location and the designated seismic event. The system **100** determines key attributes relating the key data pair. The key attributes correlate to how the parametric seismic data changes for distant geographic locations (i.e., at a distance from the event). In some embodiments, the values of key attributes are determined based principles known in physics and/or geology including, but not limited to principles of seismic attenuation, resonant vibration and/or soil behavior factors. In some embodiments, the values of key attributes are determined based on seismic attenuation factors and the distance between the property and the event center. In some embodiments, the values of key attributes are determined based on resonant vibration factors, the shaking frequency of the event, and the mass and/or resonant frequency of structures. In some embodiments, the values of key attributes are determined based on soil behavior factors and the soil at the property location, the event center, and/or at intervening geological features. Some key attributes may vary in direct (i.e., linear) proportion to the distance between the geographic location of the property/structure and the seismic event center, whereas other key attributes may vary according to reciprocal square of distance, logarithmic decay of distance or other mathematical functions relating to the distance. In some embodiments, the values of key attributes are selected by a user. Other key attributes may vary depending on other factors such as intervening geological features rather than distance. The system **100** determines, using a computer such as the system processor **102**, values of a deemed seismic action at the specified geographic location by modifying the parametric data for the designated seismic event using all relevant key attributes. In some embodiments, multiple sets of values of the key attributes are predetermined. In this case, the system processor **102** prepare a plurality of videos each combination of key attributes for each value of the key attributes.

[0111] Referring now also to FIG. **2**, there is illustrated a map **200** showing the position of the epicenter of an exemplary parametric earthquake event **202** ("Event Q"), a first exemplary geographic location **204** ("Location A") and a second exemplary geographic location **206** ("Location B"). In the map **200**, the epicenter of parametric Event Q is positioned on a geological fault line **208**; however, in other embodiments the subject event can be positioned at any location as dictated by the parametric data values. The system **100** defines a key data pair corresponding to each pair of one specified geographic location (i.e., the location of the structures/property received) and one designated seismic event. For example, Location A and Event Q form a first key data pair for showing deemed motion and/or deemed damage to properties/structures at Location A due to Event Q, and Location B and Event Q form a second key data pair for showing deemed motion and/or deemed damage to properties/structures at Location B due to Event Q. The parametric seismic data for the event **202** can be obtained from the seismic event database **112** and can include relevant parametric data for the event including, but not limited to, event name, event date, event focus location, event duration, event magnitude, event PGA and event SI.

[0112] The system **100** can determine key attributes associated with each key data pair. The key attributes can relate to factors that change the effect of the respective parametric seismic event of the key data pair on the respective geographic location of the same key data pair. As illustrated in FIG. **2**, the system **100**, e.g., using the system processor **102**, can determine a respective distance between the parametric seismic event **202** and each respective geographic location **204**, **206** associated with respective received property images and assign a respective key attribute based on the respective determined distance. For example, in FIG. **2**, the distance between the epicenter of Event Q and Location A is shown by line **210**, while the distance between Event Q and Location B is shown by line **212**. Thus, the system **100** may assign a first distance key attribute to the first key data pair (A, Q) based on the distance **210**, and assign a second distance key attribute to the second key data pair (B, Q) based on the distance **212**. In some embodiments, the distance from the geographic locations **204**, **206** to the focus (hypocenter) location of the event **202** can be used instead of the distance to the epicenter location. In some embodiments, the system **100** can detect that the path between the position of the event **202** and the geographic location **204**, **206** of a property/structure crosses an intervening geological feature **214** that can change the effects of the seismic event. In the embodiment of FIG. **2**, the path between the second key data pair (B, Q) crosses an intervening geological feature **214**. The system **100** may assign a geologic key attribute to the second key data pair (B, Q) based on a distance of the intervening geological feature **214**, whereas the first key data pair (A, Q) would not receive a geologic key attribute. In some embodiments, the system **100** can modify the parametric seismic data using all key attributes associated with the key data pair to determine the deemed seismic action and/or the deemed damage on structures at the respective geographic locations **204**, **206**. For example, the system **100** in the example of FIG. **2** can modify the parametric seismic data for event **202** in accordance with both distance key attributes and intervening geological feature key attributes (where applicable) to determine the deemed seismic action and deemed damage on structures at the respective geographic locations **204** and **206**.

[0113] The system **100** produces images showing an electronic representation(s) for the one or more structures comprising the property moving in accordance with the deemed seismic action. In some embodiments, the electronic representation(s) of the structure can be produced using a graphics animation processor **108** and an image rendering engine **110**. In some embodiments, the images comprise a video program showing an electronic representation(s) of the one or more structures moving with the deemed seismic action. In another embodiment, the images comprise an audiovisual program showing an electronic representation of the one or more structures moving with the deemed seismic action.

[0114] In some embodiments, the system **100** assigns to one or more of the identified structures a respective structure attribute. In some embodiments, the values of structure attributes are determined based principles known in physics and/or engineering including, but not limited to principles of vibrational loads, fatigue failure, strength of materials and construction methods and failure analysis. The system **100** determines values of deemed structure action for each respective structure by modifying the values of the deemed seismic action at the specified geographic location using the respective structure attribute. The system **100** produces images showing an electronic representation of the one or more structures moving in accordance with the deemed seismic action and as further modified by the deemed structure action. In some embodiments, the structure attribute is correlated to a determined mass of the structure. In other embodiments, the respective structure attribute is correlated to a determined resonant frequency of the structure. In some embodiments, the structure attribute is correlated the construction material of the structure. In some embodiments, the structure attribute is correlated the construction method type of the structure.

[0115] Referring now to FIG. **3**, there is shown an exemplary image **300** of a property disposed at a geographic location, such as first geographic location **204** or second geographic location **206**. The image **300** shows numerous man-made structures including a house **302**, a garage **304**, a driveway **306**, a pool **308**, natural structures including a yard **310** and trees **312**, and personal property items including automobiles **314**. A user of the system **100** takes the image **300** of the property and transmits it from a user device **114** to the system processor **102** via communication network **104**. In some embodiments, the image **300** may be a single image. In other embodiments, the image **300** may comprise a plurality or series of images such as a video program. The system processor **102** receives the image(s) **300** of the property and sends them an image recognition processor **106**. Using the image recognition processor **106**, the received image(s) **300** are processed to identify the one or more structures comprising the property using image recognition technology and/or artificial intelligence or machine learning technology. The system **100** can receive parametric data regarding a designated seismic event from the seismic event database **112**. The system **100** can define a key data pair corresponding to the specified geographic location of the property shown in the image **300** and the designated seismic event. The system **100** can determine, using the system processor **102**, values of a deemed seismic action at the specified geographic location of the property shown in the image **300** by modifying the parametric data for the designated seismic event using the key attributes.

[0116] Referring now to FIG. **4**, in some embodiments the system **100** can produce images **400** showing an electronic representation **402**, **404**, **406**, **408**, **410**, **412**, and **414** of the one or more structures **302**, **304**, **306**, **308**, **310**, **312** and **314** comprising the property moving in accordance with the deemed seismic action at the specified geographic location. For purposes of illustration, the representation of movement in images **400** is denoted by broken lines adjacent each structure. For example, the house **302** and garage **304** are displayed with horizontal oscillation **416** and the vertical oscillation **418** in accordance with their respective deemed seismic action. The trees **312** are shown oscillating with bending (pendulum) motion **420** in accordance with their respective deemed seismic action. The pool **308** is shown with splashing and overflowing water **422** in accordance with its respective deemed seismic action. The electronic representations **402**, **404**, **406**, **408**, **410**, **412**, and **414** of the respective moving structures **302**, **304**, **306**, **308**, **310**, **312**, and **314** can be modeled by the graphics animation processor **108** using the respective deemed seismic action and rendered by the image rendering engine **110** based on the respective structures identified in the original image **300**.

[0117] After producing the image of the electronic representations **400**, **402**, **404**, **406**, **408**, **410**, **412**, and **414** of the one or more structures **302**, **304**, **306**, **308**, **310**, **312**, and **314** comprising the property moving in accordance with the deemed seismic action at the specified geographic location, the system **100** can transmit the images **400** back to the first user device **114** or to another user device **115** designated by the original request. The user can then view the electronic representation **400** using a display device **120**.

[0118] In another aspect, a system **100** can receive images of property at a specified geographic location, processes, using an image recognition processor **106**, the received images, identify one or more structures comprising the property and receive parametric data regarding a designated seismic event, all substantially as previously described. The system **100** can further define a key data pair corresponding to the specified geographic location and the designated seismic event, determine key attributes relating the key data pair, and determine, using a computer, e.g., processor **102**, values of a deemed seismic action at the specified geographic location by modifying the parametric data for the designated seismic event using the key attributes, again, all substantially as previously described. The system **100** can further determine a respective deemed damage corresponding to each structure based on the deemed seismic action. The system **100** produces images showing an electronic representation the one or more structures comprising the property modified with respective electronic representations of the respective deemed damage.

[0119] In some embodiments, the structure attribute used to determine the deemed damage is correlated to a determined mass of the structure. In other embodiments, the structure attribute is correlated to a determined resonant frequency of the structure. In still other embodiments, the structure attribute is correlated to a construction material of the structure. Thus, structures built with shake-resistant materials or building methods would have a different structure attribute than structures built with shake-damage prone materials or building methods.

[0120] Referring now to FIG. **5**, in one example, the system **100** can produce images **500** showing an electronic representation **400**, **402**, **404**, **406**, **408**, **410**, **412**, and **414** the one or more structures **302**, **304**, **306**, **308**, **310**, **312** and **314** comprising the property modified with respective electronic representations of the respective deemed damage. For example, the original appearance of the house **302** from image **300** is modified with electronic representations of cracked windows **502**, wall cracks **504** and structural damage **506**. The original appearance of the garage **304** from image **300** is modified with electronic representations of structural collapse **508** and damaged automobiles. The original appearance of the driveway **306** from image **300** is modified with electronic representations of cracking **509**. The original appearance of the pool **308** from image **300** is modified with electronic representations of cracks **510** on the shell and decking. The original appearance of the trees **312** from image **300** is modified with electronic repositioning to represent leaning or toppled **512**. The original appearance of the yard **310** from image **300** is modified with electronic representations of washouts **514** near the pool **308** and landslide/subsidence **516**. cracks **510** on the shell and decking.

[0121] After producing image **500** the electronic representation **400**, **402**, **404**, **406**, **408**, **410**, **412**, and **414** of the one or more structures comprising the property modified to show the respective deemed damage associated with a parametric seismic event, the system **100** can transmit the images **500** back to the original user device **114**′ or to another user device **114**″ designated by the original request. The user can then view the electronic representation **400** using a display device **120**.

[0122] The system processor **102** can include any suitable hardware processor, such as a microprocessor, and in some embodiments, the hardware processor can be controlled by a program stored in the memory and/or storage. The image recognition processor **106**, graphics animation processor **108** and image rendering engine **110** can include any suitable hardware, and each can be optimized for graphics-intensive computing with the incorporation of one or more graphics processing units (GPUs). Communication interfaces for the processors **102**, **106**, **108** and rendering engine **110** can be any suitable network communication interface, but can be optimized in some embodiments for high speed data transfer between the graphics processing devices **106**, **108** and **110**.

[0123] Referring now to FIGS. **6-10**B, methods are illustrated methods of modifying the images of respective original structures to produce an electronic representation of the one or more structures modified to show the respective deemed damage. FIG. **6** shows an exemplary image **600** of actual roofing **602** of a structure, such as might be included in an image **300** received from a user of the system **100**. The image recognition processor **106** of the system **100** can identify the image **600** as showing “roofing,” therefore the graphics animation processor can select image textures corresponding to “damaged roofing.” FIG. **7** shows an electronic representation **700** of an exemplary damage texture **702** representing “damaged roofing,” which texture can be an actual photo of damaged roofing or a digital representation thereof. FIG. **8**A shows an exemplary first texture map **800** with a randomized distribution of damage areas **802** corresponding in density to a first amount of deemed damage. In some embodiments, the shapes of the damage areas **802** included in a damage texture may be arbitrary shapes or randomly generated shapes. In some embodiments, the shapes of the damage areas **802** included in a damage texture can be predetermined based on representative types of damage. FIG. **8**B shows the first damage texture **702** applied to the first texture map **800** such that the damage texture appears in the randomized damage areas **802**′ to produce an electronic representation **810** of a first level of deemed damage. FIG. **9**A shows an exemplary second texture map **900** with a randomized distribution of damage areas **902** corresponding in density (in this case, a higher density) to a second amount (in this case, a higher amount) of deemed damage. FIG. **9**B shows the first damage texture **702** applied to the second texture map **900** such that the damage texture appears in the randomized damage areas **802**″ to produce an electronic representation **910** of the second, higher, level of deemed damage with a greater percentage of the texture representing damage texture **702**.

[0124] FIG. **10**A shows the original image **600** of the structure roofing of FIG. **6** modified using the electronic representation **810** of the first level of deemed damage to produce an electronic representation **1000** of the structure damage **1002** in accordance with the first level of deemed damage. FIG. **10**B shows the original image **600** of the structure roofing of FIG. **6** modified using the electronic representation **910** of the second first level of deemed damage to produce an electronic representation **1005** of high-level structure damage **1007** in accordance with the second level of deemed damage.

[0125] Referring now to FIGS. **11-14**, additional methods are shown for modifying the images of respective original structures to produce an electronic representation the one or more structures modified with respective electronic representations of the respective deemed damage. FIG. **11** shows an image **1100** of an actual structure **1102** on a property, in this case a house. The image recognition processor **106** of the system **100** can identify the image **1100** as showing a “house.” In some embodiments, one of the image recognition processor **106** and the graphics animation processor **108** can recognize the substructures of a typical house or actual structure **1102** to identify walls **1104**, windows **1106** and/or roof **1108**.

[0126] Referring now to FIG. **12**, after processing of the original image **1100** by the image recognition processor **106**, and optionally, the identification of substructures **1104**, **1106** and **1108**, the system **100** can produce a wireframe representation **1200** of the actual structure **1102**. In some embodiments the wireframe representation **1200** is produced by the graphics animation processor **108** and in other embodiments it is produced by other parts of the system **100**. The wireframe representation **1200** can include coordinate locations **1202** corresponding to portions of the actual structure **1102**. In the example of FIG. **12**, the coordinate locations **1202** correspond to vertices of the roof **1108** in the original image **1100**.

[0127] As previously described, the system **100** can determine a respective deemed damage corresponding to each respective structure based on the deemed seismic action. FIG. **13** shows an electronic representation **1300** of a damaged wireframe produced by modifying the original wireframe representation **1200** of FIG. **12** in accordance with an assigned level of deemed damage received from the system processor **102**. In some embodiments, the damaged wireframe **1300** is produced by displacing coordinate locations **1202** from their original location **1302** (shown in broken line) in wireframe **1200** to shifted locations **1304** (shown in solid), wherein the displacement between the original location **1302** and shifted locations **1304** is correlated to the assigned level of deemed damage. In the example of FIG. **13**, only the roof structure is shown modified; in other embodiments additional coordinate locations, e.g., for walls, windows, etc., can also be displaced.

[0128] After modifying the wireframe representation **1200** of the original image to produce the damaged wireframe representation **1300**, the system **100** can produce an electronic representation of the structure in accordance with the deemed damage level by rendering texture on the damaged wireframe representation. In some embodiments, the texture rendering for the electronic representation can be performed by the image rendering engine **110**. Further, in some embodiments, the electronic representation of the structure can be further modified by replacing the original texture with an electronic representation of damage in the texture in accordance with the assigned level of deemed damaged (e.g., as described in connection with FIGS. **6-10**B. For example, FIG. **14** shows an electronic representation structure **1400** of the structure **1102** of FIG.

11 damaged in accordance with the deemed damage level by rendering texture on the damaged wireframe representation to represent areas of structural damage **1402** and modifying the original texture with an electronic representation of damaged texture **1404** in accordance with the assigned level of deemed damaged.

[0129] Referring now to FIGS. **15**A-**15**C and FIGS. **16**A-**16**E, there is illustrated a system **1500** for display of an electronic representation of physical effects and property damage resulting from a parametric earthquake event (hereinafter “Earthquake Visualization” system or “EQV” system) in accordance with another aspect. The EQV system **1500** can include hardware such as that in system **100** of FIG. **1**, e.g., system server/processor **102**, image recognition processor **106**, graphics animation processor **108**, and image rendering engine **110**. The EQV system **1500** can include some apparatus, software and processes performed at a user location and other apparatus, software and processes performed at a remote location, wherein the user location and remote location are operably connected by a digital communication network, (e.g., communication network **104** of FIG. **1**). In the illustrated embodiment, the EQV apparatus, software and processes performed at the user location are shown in FIGS. **15**A-**15**C, the EQV apparatus, software and processes performed at the remote location are shown in FIGS. **16**A-**16**E, and the interconnecting digital communication network is a global communication network, for example the internet. In other embodiments, the disclosed EQV apparatus, software and processes may be distributed differently between the user location and the remote location. In still other embodiments, the disclosed EQV apparatus, software and processes may be distributed between the user location and several remote locations. In addition to the internet, alternative digital communication networks can be used with the EQV system **1500** as long as they enable digital communications between the user location and the remote location.

[0130] Referring now specifically to FIG. **15**A, a user initiates the EQV system **1500** on a computing device at the user location. The computing device can be a mobile communication device including, but not limited to, a smart phone, tablet, smart watch, a computer including, but not limited to, a desktop or laptop computer, or a home automation device including, but not limited to, an Amazon® Echo® or Google® Home® device. The user initiates the EQV system **1500** by downloading application software onto the computing device.

[0131] Summarizing some of the methods shown in FIGS. **15**A-**15**C, with reference to selected blocks of the diagram (as denoted): (Block **1502**) User initiates EQV App on mobile device. Optionally, EQV App determines user's location using GPS. Optionally, EQV App gets user info. and contact information.

[0132] User selects type of video desired in EQV App, including, but not limited to: Inside of structure; outside of structure; yard; garage with car(s); pool (Block **1506**). Optionally, user takes video of property with mobile device camera using EQV App (Block **1508**). Additionally, a user can upload existing videos (Block **1510**). Video is uploaded to EQV server/processor **1514** using EQV App (Block **1512**).

[0133] The process continues as shown in FIG. **15**B (block **1516**). The user selects SI/PGA using the EQV App (Block **1518**). Optionally, user specified actual value. Optionally, user selects “historical value” for SI/PGA equivalent to values for named historical earthquake event. SI/PGA uploaded to EQV server/processor **1514** using EQV App (Block **1520**).

[0134] The user can select classifications or identify specific objects, such as a type of car in a garage (Block **1522**). The classifications or identified specific objects can be uploaded to the EQV server/processor **1514** (Block **1524**). The user can select scene dimensions and other characteristics, such as property square feet and/or value of a property (Block **1526**). The scene dimensions and other characteristics can be uploaded to the EQV server/processor **1514** (Block **1528**). The user can select additional scenario parameters (Block **1530**). The additional scenario parameters can be uploaded to the EQV server/processor **1514** (Block **1532**).

[0135] The process continues as shown in FIG. **15**C (Block **1534**). The user can initiate scenario processing (Block **1536**) at the EQV server/processor **1538**. EQV server/processor **1538** can be the same as or different from EQV server/processor **1514**. The user device can receive processed video(s) from the EQV server/processor **1538** (Block **1540**). The user device can receive real property damage prediction(s) from the EQV server/processor **1538** (Block **1542**). The user device can receive personal property damage prediction from the EQV server/processor **1538** (Block **1544**).

[0136] The user device can receive a share link for one or more seismic damage video(s) (Block **1546**). The user device can play the one or more seismic damage video(s) and views damage prediction (Block **1548**). The user device can download the one or more video(s) to save and share to others using the share link (Block **1550**).

[0137] Summarizing some of the methods shown in FIGS. **16**A-**16**E with reference to selected blocks of the diagram **1600** (as denoted): EQV server/processor is initiated (Block **1602**) and EQV server/processor processes uploaded video according to designated SI/PGA (Block **1604**). The user then designates the type of video uploaded (Blocks **1606**, **1622**, **1640**, **1656**). In other embodiments, the type of video uploaded can be determined by the system, e.g., using image processing and/or machine learning.

[0138] Home interior video selected (Block **1606**): Physics-based motion effects applied to video image of the interior structure during simulated event (Block **1608**). Physics-based cracking effects applied to image of walls (Block **1610**). Physics-based breakage effects applied to image of windows (Block **1612**). Physics-based damage applied to image of ceilings (e.g., full or partial collapse) (Block **1614**). Physics-based movement effects applied to image of furnishings (Block **1616**).

[0139] The diagram continues in FIG. **16**B (Blocks **1618** and **1620**). Home exterior video selected (Block **1622**): Physics-based motion effects applied to image of the exterior structure during simulated event (Block **1624**). Physics-based motion effects applied to images of trees and vegetation during event (e.g., dropping leaves; falling limbs; falling trees) (Block **1626**). Physics-based structural failure applied to image of bricks, chimneys, fences (Block **1628**). Physics-based cracking effects applied to image of walls (Block **1630**). Physics-based breakage effects applied to image of windows (Block **1632**). Physics-based damage effects applied to image of roof (e.g., full or partial collapse) (Block **1634**). Physics-based earth movement effects applied according to designated SI/PGA (e.g., landslide; house movement on slope, etc.) (Block **1636**).

[0140] The diagram continues in FIG. **16**C (Block **1638**). Garage video can be selected (Block **1640**) the same as the home interior, but add physics-based damage effects to structure images during event duration (Block **1642**). Physics-based cracking effects can be applied to wall images (Block **1644**). Physics-based breakage effects can be applied to window images (Block **1646**). Physics-based damage effects can be applied to ceiling images (Block **1648**). Physics-based movement effects can be applied to furnishing images (Block **1650**). Physics-based damage effects applied to image of car(s) from structural collapse of garage (**1652**).

[0141] The diagram continues in FIG. **16**D (Block **1654**). Pool video can be selected (Block **1656**): Physics-based water slosh effect added to water surface during simulated event (Block **1658**). Physics-based cracking effects applied to image of walls (Block **1660**). Physic-based cracking effects can be applied to decks (Block **1662**).

[0142] Damage Predictors (optional): Real Property Damage Prediction can be performed (Block **1664**); EQC Server/Processor determines approximate square feet of house and/or value of house from data source (public/private) based on GPS location from EQV App. (Alternative) User provides estimated real property value via EQV App. EQV Server/Processor calculated predicted value of real property damage for user-specified SI/PGA. Personal Property Damage Prediction can be performed (Block **1666**). A share link can be generated for the videos (Block **1668**).

[0143] The diagram continues in FIG. **16**E (Block **1670**). As previously described, an image labeling or annotation tool may be incorporated into the system to detect, identify, recognize, and/or mark objects in the video and/or still images. The EQV Server/Processor counts personal property objects recognized in video (while adding effects), e.g., number lamps, number of paintings, number of tables, number of chairs, cars in garage (if applicable), etc. (as many labels as desired). EQV Server/Processor sets value to each labeled object based on database of values. (Alternative) User provides estimated personal property value via EQV App. EQV Server/Processor calculates predicted value of personal property damage for user-specified SI/PGA.

[0144] EQV server/processor downloads processed video to user device **1680** on EQV App (Block **1672**). (Optional) EQV server/processor downloads predicted value of real property damage (Block **1674**) and/or personal property damage (Block **1676**) to the user device **1680**. The EQV server/processor can download a share link (Block **1678**) to the user device **1680**.

[0145] After receiving the processed video from the EQV server/processor, the user at user device **114** plays video or views still photos showing electronic representation of structure motion and/or structure damage in accordance with the parametric seismic event and the geographic location of the property comprising the structures. Optionally, the user views predicted damage values. User selects new video or new SI/PGA. Repeat process if desired with different images, different locations, different event types and/or different event parameters.

[0146] Summarizing some of the methods relating to Effects Modeling in accordance with additional aspects: Motion (shake) effects added to entire video image based on designated SI/PGA (i.e., speed and magnitude are appropriate for PGA). Damage effects applied based on polygon modeling, texture sampling and texture substitution. Object in video is selected. Object is digitized into original polygon model (shape) and sampled for original texture (color and pattern). Object's original polygon model is deformed to create damaged polygon model; amount of deformation is based on designated SI/PGA. Object's original texture is replaced by damaged texture comprising original texture interspersed with contrasting “feature texture” (e.g., can be specific to type of damage such as cracks, exposed lumber, missing shingles, etc.); proportion of feature texture mixed to original texture is based on designated SI/PGA. Damaged object image is formed from damaged polygon model covered with damaged texture. Original object image is then overlayed by damaged object image in video.

[0147] Referring now to FIGS. **17**, in some embodiments a structure, such as house **302** or garage **304** shown in FIG. **3**, can include an interior room such as that pictured in interior image **1700**. The interior image **1700** can be a still image or an image from a video. The interior image **1700** can capture a scene that includes structural components including one or more walls, such a first wall **1702**, second wall **1704** including one or more doors **1706**, and third wall **1708** including one or more windows **1710**, a floor **1712**, a ceiling, or any other suitable structural component. The scene of the interior image **1700** can also include personal items, including a television **1714** mounted on a wall (such as the first wall **1702**), a picture **1716** mounted on a wall (such as the first wall **1702**), a coffee table **1718** at a center of the floor **1712**, a couch **1720** on the floor **1712** to a side of the coffee table **1718**, a table **1722** against the third wall **1708** under the window **1710**, a candle **1724** on the coffee table **1718**, a vase **1726** on the table **1722**, and any other suitable personal items.

[0148] Referring now to FIG. **18**A, there is shown an exemplary user interface **1800**. The first user device **114** can use the camera **116** to capture an interior image **1700** of the scene shown in FIG. **17**. Alternatively, the interior image **100** can be retrieved from storage. The interior image **1700** is uploaded into the system either on a device or in a remote location. The user interface **1800** can be displayed on the display device **120** of the first user device **114**. The system can process the interior image **1700** to generate digital representations **1802-1826** for each of the objects **1702-1726** shown in FIG. **17** for display on the user interface **1800**. The generation of the digital representations **1802-1826** can occur in the user device **114** or in a server **102** that the user device **114** communicates with. The user interface **1800** can further include a damage selection window **1828** for selecting a type of damaging event to visualize in the user interface **1800**. The damage selection window **1828** can include a flood input **1830**, a seismic input **1832**, a wind input **1834**, or any other suitable input **1836**. Once an input has been selected, additional windows related to the specific event can also be displayed. In certain embodiments, the additional windows are always displayed and the values are changed based on the selection of the damage selection window **1828**.

[0149] Referring now to FIG. **18**B, the flood input **1830** has been selected on the damage selection window **1828** of the user interface **1800**. Once the flood input **1830** is selected from the damage selection window **1828**, a flood time window **1838** and water level selection window **1840** are displayed. The flood time selection window **1838** controls the time of the flooding event displayed on the user interface. The flood time selection window **1838** can have predetermined times set, such as a "during" input **1842** for showing a representation of the room while the flooding event is in progress, an "after" input **1844** for showing a representation of the room after the flooding event has ended but the water remains (shown selected in FIG. **18**B), a "drain" input **1846** for showing a representation of the room after the water has been drained (shown in FIG. **18**C), and time selection input **1848** for entering an intermediate time during the flooding event. The water level selection window **1840** can provide different predetermined levels of flood intensity including a first level input **1850** (shown selected in FIGS. **18**B and **18**C), a second level input **1852**, a third level input **1854**, "enter value" level input **1856**, or any other suitable intensity level inputs. The predetermined levels can be based on height of water for historic storms, be a percentage of the room, or any other suitable way to determine a water level.

[0150] As shown in FIG. **18**B, the flood input **1830** is selected on the damage selection window **1828**, the after input **1844** is selected on the flood time selection window **1838**, and the first level input **1850** is selected on the water level selection window **1840**. The represented water level **1858** corresponds to the first level input **1850**. The represented water level **1858** rises up the digital representation walls **1802**, **1804**, **1808**, covers digital representation coffee table **1818**, and partially covers the digital representation couch **1820** and the digital representation table **1822**. As shown in FIG. **18**C, the selection in the flood time selection window **1838** is changed from the after input **1844** to the drained input **1846**. The water has been drained in the user interface **1800**. The represented digital representation walls **1802**, **1804**, and **1808** now show digital representation mold **1860** or other damage. The digital representation couch **1820** can be made of a fabric and show digital representation stain(s) **1861**. The digital representation table **1822** could be made of wood and show digital representation rot **1862**.

[0151] Referring now to FIG. **18**D, the seismic input **1832** has been selected on the damage selection window **1828** of the user interface **1800**. Once the seismic input **1832** is selected from the damage selection window **1828**, a seismic time window **1864** and seismic intensity window **1866** are displayed. The seismic time window **1864** controls the time of the seismic event displayed on the user interface. The seismic time window **1864** can have predetermined times set, such as during input **1842** showing the room representation with the seismic event is in progress, an after input **1844** for showing the room representation after the seismic event has ended (shown selected in FIG. **18**D), and time selection input **1848** for entering an intermediate time during the seismic event. The seismic intensity window **1866** can provide different predetermined levels of represented seismic intensity including a first level, a second level **1852**, a third level **1854** (shown selected in FIG. **18**D), an enter value level **1856**, or any other suitable intensity levels. The predetermined levels can be based on strength of an earthquake for historic quake events or any other suitable way to determine a seismic level or massive vibration level.

[0152] The user interface **1800**, shown in FIG. **18**D, displays a representation of seismic damage that would be cause by an earthquake or other massive vibration event. The seismic damage caused a digital representation crack **1868** in the digital representation first wall **1802** from the ceiling or top of the digital representation first wall **1802**. The digital representation picture **1816** is crooked, the digital representation television **1814** has fallen off the digital representation first wall **1802** and includes a digital representation crack **1870** in the screen. The vibrations caused the digital representation vase **1826** to fall and break into digital representation pieces **1872**. The digital representation candle **1824** fell off the digital representation coffee table **1818**. The digital representation couch **1820** has shifted.

[0153] Referring now to FIG. **18**E, the wind input **1834** has been selected on the damage selection window **1882** of the user interface **1800**. Once the wind input **1834** is selected from the damage selection window **1828**, a wind time window **1874** and wind speed selection window **1876** are displayed. The wind time window **1874** controls the time of the represented wind event displayed on the user interface. The wind time window **1874** can have predetermined times set, such as a during input **1842** the wind event is in progress, an after input **1844** for after the wind event has ended (shown selected in FIG. **18**E), and time selection input **1848** for entering an intermediate time during the wind event. The wind speed selection window **1876** can provide different predetermined levels of represented wind intensity including a first level input **1850**, a second level input **1852**, a third level input **1854**, enter value level input **1856** (shown in FIG. **18**E), or any other suitable intensity levels. The predetermined levels can be based on strength of wind or category of historic storms or any other suitable way to determine a wind strength level.

[0154] As shown in FIG. **18**E, the after input **1844** is selected such that the user interface **1800** displays the represented aftereffects of a wind damaging event. A specific value input **1856** has been entered in the wind speed selection window **1876**. The value selected by the user was strong enough to shatter **1877** the digital representation window **1810** on the digital representation third wall **1808**. The represented wind caused the digital representation picture **1816** to rotate and the digital representation couch **1820** to move back to the second wall **1804** blocking the digital representation door **1806**. The represented wind strength knocked the digital representation candle **1824** off of the digital representation coffee table **1818**. The represented wind also knocked the digital representation vase **1826** off of the digital representation table **1822** and break into digital representation pieces **1872**.

[0155] As shown in FIGS. **18**A-**18**E, the user interface **1800** can also include a destructive event control bar **1878**. The destructive event control bar **1878** controls a timeline of the destructive event in the user interface **1800**. The destructive event control bar **1878** can include a destructive event time indicator **1880** with a destructive event progress indicator **1882**, a destructive event play button **1884**, and a destructive event pause button **1886**. The destructive event time indicator **1880** shows a total amount of time that the destructive event occurs. This could include amount of time for waters to recede and mold to form on the digital representation walls **1802**, **1804**, **1808** during a flooding event. The destructive event progress indicator **1882** indicates the current time in the of the destructive event displayed on the user interface **1800**. The destructive event progress indicator **1882** can be selectively adjusted to adjust the video of the destructive event on the user interface **1800**. The destructive event play button **1884** can start the video at the current time of the destructive event. The destructive event pause button **1886** can pause the video at the time of the destructive event.

[0156] FIG. **19** is a block diagram of a system for display of an electronic representation of physical effects and property damage resulting from a parametric earthquake event in accordance with various embodiments of the disclosure. As shown in FIG. **19**, there is illustrated a system **1900** for display of an electronic representation of physical effects and property damage resulting from a parametric earthquake event in accordance with one aspect. Unless otherwise noted, in this application the terms “parametric earthquake event” and “parametric seismic event” have the same meaning. A parametric earthquake event is a set of parameters and values for defining an earthquake event in terms of scientific principles, e.g., physics and geological principles. Such parameters can include, but are not limited to, date, time, epicenter location, hypocenter location (i.e., focus), duration, magnitude, peak ground acceleration (“PGA”), and maximum shaking intensity (“SI”), maximum ground amplitude, mean ground amplitude, shaking frequency, soil type, rock type and fault classification. A parametric earthquake event can include some or all parameter values measured during an actual, i.e., “historical” earthquake event, but can also include some or all parameters values selected by a user. The system **1900** includes a system server **1902** or computing device. The system server **1902** contains one or more processors, RAM memory, storage units and communication interfaces for sending and receiving data to and from other devices. The system server **1902** may comprise a single machine or multiple machines, including virtual machines resident on “cloud servers.” The system server **1902** is operably connected to a communication network **1904** and to one or more subsystems, e.g., an image recognition processor **1906**, a graphics animation processor **1908** and/or an image rendering engine **1910**. In some embodiments, the communication network **1904** is the internet; however, the communication network **1904** can be any type of network that allows the system server **1902** and subsystems **1906**, **1908** and **1910** to communicate with one another and with a user device **1914**. In the embodiment illustrated in FIG. **19**, the server **1902** is directly connected to the subsystems **1906**, **1908** and **1910**; however, in other embodiments some or all of the subsystems may be connected to the server via the communication network **1904**. In some embodiments, the subsystems **1906**, **1908** and **1910** can communicate directly with one another for improved data transfer.

[0157] The image recognition processor **1906** can include or incorporate an image labeling or annotation tool (i.e., “labeling tool”). The labeling tool can be used to process and label the images for bounding box object detection and segmentation so that the image is readable by machines. In some embodiments, the labeling tool can utilize human assistance and in other embodiments the labeling tool can operate solely with machine learning or artificial intelligence processes. In some embodiments, different image labeling tools may be used for processing images of different image types (e.g., interior images, exterior images, etc.). Using the image labeling tools, the various objects in the provided image can be labeled for specific purposes. In some embodiments, labeled objects can be selected for replacement by a computer graphic object (e.g., 2D sprite or 3D polygon), which can be moved on-screen and otherwise manipulated as a single entity, e.g., for purposes of event animation. In some embodiments, labeled objects may be classified into different types or categories of objects for different purposes. For example, in some embodiments, labeled objects can be categorized for properties relating to event animation, e.g., movable-type objects, bendable-type objects, breakable-type objects, waterproof-type objects, water damageable-type objects, etc. In other embodiments, labeled objects can be categorized for properties relating to inventory or damage assessment, e.g., table-type objects, chair-type objects, window-type objects, door-type objects, hanging art-type objects, TV/computer screen-type objects, etc. Such classification may be performed by the image recognition processor **1906** or by another processor, e.g., the system processor **1902** or graphics animation processor **1908**. The image labeling tools can use known or future-developed detection techniques for detection of the object including, but not limited to, semantic, bounding box, key-point and cuboid techniques.

[0158] A seismic event database **1912** is operably connected to the server **1902**, either via the communication network **1904** or directly. The seismic event database **1912** stores parametric seismic event data corresponding to one or more earthquakes or other seismic events. The parametric seismic event data can include, but is not limited to, values for the following parameters: event name, event date, event epicenter location, event focus (i.e., hypocenter) location, event duration, event magnitude, event PGA, event maximum shaking intensity (“SI”), event maximum ground amplitude, event mean ground amplitude, and event shaking frequency for each seismic event. The seismic events data in the seismic event database **1912** may be actual historical earthquake data, “relocated” earthquake data (i.e., where the majority of the data corresponds to a historical earthquake, but the epicenter/hypocenter location is changed to a different location) or hypothetical earthquake data specified by a user or otherwise generated. In some embodiments, the seismic event database **1912** can be located within the server **1902** or one of the subsystems. In some embodiments, the seismic event database **1912** can include data and/or values from public or private earthquake reporting agencies, such as the U.S. Geological Survey.

[0159] User devices **1914**, **1915** can connect to the system **1900** through the communication network **1904**. The user devices **1914**, **1915** can be mobile devices such as mobile phones, tablets, laptop computers or they can be stationary devices such as desktop computers or smart appliances including, but not limited to, smart televisions. In some embodiments, aspects of the current system **1900** may include downloadable software or "apps" resident on the user devices **1914**, **1915** and/or non-downloadable software (e.g., "cloud based software") that remains resident on the server **1902** or other elements of the system **1900** and is accessed by the user device **1914** via a web browser or other network interface.
[0160] Using the user devices **1914**, **1915**, system users can upload images of actual property via the network **1904** to the server **1902**. The images of actual property can be captured using a camera **1916** on the first user device **1914** or from images stored in memory **1918**, e.g., a computer memory, hard drive, flash drive or other data storage technology. The images can be video programs, audiovisual programs and/or single or multiple still photo images. User devices **1914** can also transmit additional information to the server **1902** regarding the images, including, but not limited to, the geographic location of the property in the images (either entered by the user as text, captured via GPS or wireless location information on the user device **1914**, or captured vis geotagging information on the image file), the address of the property in the images, the name and/or other contact information of the user, a desired parametric seismic event and/or desired seismic parameters to be used in creating a parametric seismic event. Desired seismic parameters can include strength of the seismic event, duration of the seismic event, etc. Each of the seismic parameters can be input on the user devices **1914**, **1915** by any suitable means. For example, the seismic parameters can be input or selected using a predetermined list, a slider associated with different values of the seismic parameters, a knob, a number entry. The inputs can be physical components on the user device **1914**, **1915** or virtual representations on a display **1920** of the user device **1914**, **1915**.
[0161] The system **1900** processes, using an image recognition processor **1906**, the received images and identifies one or more structures comprising the property. The system **1900** retrieves parametric data regarding a designated seismic event from the seismic event database **1912**. The system **1900** defines a key data pair corresponding to the specified geographic location and the designated seismic event. The system **1900** determines key attributes relating the key data pair. The key attributes correlate to how the parametric seismic data changes for distant geographic locations (i.e., at a distance from the event). In some embodiments, the values of key attributes are determined based principles known in physics and/or geology including, but not limited to principles of seismic attenuation, resonant vibration and/or soil behavior factors. In some embodiments, the values of key attributes are determined based on seismic attenuation factors and the distance between the property and the event center. In some embodiments, the values of key attributes are determined based on resonant vibration factors, the shaking frequency of the event, and the mass and/or resonant frequency of structures. In some embodiments, the values of key attributes are determined based on soil behavior factors and the soil at the property location, the event center, and/or at intervening geological features. Some key attributes may vary in direct (i.e., linear) proportion to the distance between the geographic location of the property/structure and the seismic event center, whereas other key attributes may vary according to reciprocal square of distance, logarithmic decay of distance or other mathematical functions relating to the distance. In some embodiments, the values of key attributes are selected by a user. Other key attributes may vary depending on other factors such as intervening geological features rather than distance. The system **1900** determines, using a computer such as the system processor **1902**, values of a deemed seismic action at the specified geographic location by modifying the parametric data for the designated seismic event using all relevant key attributes. In some embodiments, multiple sets of values of the key attributes are predetermined. In this case, the system processor **1902** prepares a plurality of videos each combination of key attributes for each value of the key attributes.

[0162] In certain embodiments, the images of actual property can be captured using a camera **1916** on the first user device **1914** or from images stored in memory **1918**, e.g., a computer memory, hard drive, flash drive or other data storage technology. User devices **1914** can also store additional information regarding the images, including, but not limited to, the geographic location of the property in the images (either entered by the user as text, captured via GPS or wireless location information on the user device **1914**, or captured vis geotagging information on the image file), the address of the property in the images, the name and/or other contact information of the user, a desired parametric seismic event and/or desired seismic parameters to be used in creating a parametric seismic event. Desired seismic parameters can include strength of the seismic event, duration of the seismic event, etc. Each of the seismic parameters can be input on the user devices **1914**, **1915** by any suitable means. For example, the seismic parameters can be input or selected using a predetermined list, a slider associated with different values of the seismic parameters, a knob, a number entry. The inputs can be physical components on the user device **1914**, **1915** or virtual representations on a display **1920** of the user device **1914**, **1915**.
[0163] The processor **1922** on the user devices **1914**, **1915** can process the received images and identifies one or more structures comprising the property. The processor **1922** can retrieve parametric data regarding a designated seismic event from the seismic event database **1912** via the communication network **1904**. The processor **1922** can define a key data pair corresponding to the specified geographic location and the designated seismic event. The processor **1922** can determine key attributes relating the key data pair. The key attributes correlate to how the parametric seismic data changes for distant geographic locations (i.e., at a distance from the event). In some embodiments, the values of key attributes are determined based principles known in physics and/or geology including, but not limited to principles of seismic attenuation, resonant vibration and/or soil behavior factors. In some embodiments, the values of key attributes are determined based on seismic attenuation factors and the distance between the property and the event center. In some embodiments, the values of key attributes are determined based on resonant vibration factors, the shaking frequency of the event, and the mass and/or resonant frequency of structures. In some embodiments, the values of key attributes are determined based on soil behavior factors and the soil at the property location, the event center, and/or at intervening geological features. Some key attributes may vary in direct (i.e., linear) proportion to the distance between the geographic location of the property/structure and the seismic event center, whereas other key attributes may vary according to reciprocal square of distance, logarithmic decay of distance or other mathematical functions relating to the distance. In some embodiments, the values of key attributes are selected by a user. Other key attributes may vary depending on other factors such as intervening geological features rather than distance. The processor **1922** can determine values of a deemed seismic action at the specified geographic location by modifying the parametric data for the designated seismic event using all relevant key attributes. In some embodiments, multiple sets of values of the key attributes are predetermined. In this case, the processor **1922** can prepare a plurality of videos each combination of key attributes for each value of the key attributes.
[0164] The user device **1914**, **1915** can also include a storage device **1924**. A memory and a persistent storage are examples of storage devices **1924**, which represent any structure(s) capable of storing and facilitating retrieval of information (such as data, program code, and/or other suitable information on a temporary or permanent basis). The storage device **1924** may represent a random access memory or any other suitable volatile or non-volatile storage device(s). The storage device **1924** may contain one or more components or devices supporting longer-term storage of data, such as a read only memory, hard drive, Flash memory, or optical disc.
[0165] The storage device **1924** can also include an AR plane manager application **1926**, an AR mesh manager application **1928**, and an AR camera manager application **1930**. The processor **1922** can run the AR plane manager application **1926**, the AR mesh manager application **1928**, and the AR camera manager application **1930** for display or an electronic representation of physical effects

and property damage resulting frame a parametric natural disaster event.
[0166] The AR plane manager application **1926**, run using the processor **1922**, can extract AR planes from a scene. The AR planes can be 3D planes that can be defined by a position, an orientation, and bounds. The AR planes can be rendered on a display **1920**. The AR planes can also be classified into categories, such as wall, floor, ceiling, door, window, seat, table, none, etc.
[0167] The AR mesh manager application **1928** can mesh the Lidar point cloud in an AR wrapper. For example, the AR wrapper can be AR Foundation provided by Unity on platforms such as Apple iOS, such as iOS system ARKIT for Visual Inertial Odometry and 3D meshing, and Google Android. The 3D mesh consists of vertices, triangle definitions, and surface normal, as shown in FIG. **22**. The 3D mesh can be organized in a form of regions. Each region can contain a 3D mesh, where the 3D mesh can be managed and updated independently from other 3D meshes.
[0168] The AR camera manager application **1930** can provide a tracked six degree of freedom pose of a camera **1916** at every frame. The AR camera manager application **1930** can extract a camera image and camera parameters at each frame for computer vision applications.
[0169] Although FIG. **19** illustrates an example of a system for display of an electronic representation of physical effects and property damage resulting from a parametric earthquake, various changes may be made to FIG. **19**. For example, various components in FIG. **19** may be combined, further subdivided, replicated, omitted, or rearranged and additional components may be added according to particular needs.
[0170] Specifically, in some embodiments, the system for display of an electronic representation of physical effects and property damage resulting from a parametric earthquake event can be provided in a stand-alone mobile device such as, but not limited to, a mobile phone, tablet or similar consumer electronic device. In such embodiments, the user device **1914** itself incudes the processor **1922**, display device **1920**, input device (e.g., touch screen), camera **1916** (including optical cameras and, optionally, Lidar or other distance sensors), image storage **1918** and storage device **1924**. In such mobile device-based embodiments, the device **1914** may also include an image recognition processor **1906**, graphics animation processor **1908** and/or image rendering engine **1910** implemented in hardware or software on the mobile device instead of, or in addition to, processors accessed remotely through a network **1904** and server **1902**. In such mobile device-based embodiments, the device **1914** may also include a seismic event database **1912** stored in the memory device instead of, or in addition to, a seismic event database accessible via the network.
[0171] FIGS. **20**A and **20**B illustrate an example method for an AR earthquake effect visualization and assessment process **2000** in accordance with this disclosure. As shown in FIGS. **20**A and **20**B, the AR earthquake effect visualization process **2000** is provided to produce a visual representation of possible damage that would be caused in an earthquake. The earthquake effect visualization process can include a scene generation subprocess **2002** shown in FIG. **20**A and a simulation sub-process **2004** shown in FIG. **20**B.
[0172] As shown in the scene generation subprocess **2002** of FIG. **20**A, the processor **1922** can scan a scene using an optical recognition method in operation **2006**. In certain embodiments, the scene can be an interior of a room and one or more walls of the room can be scanned. The user device **1914** can further include a time-of-flight camera **1916** or Lidar sensor, which the processor **1922** can operate to scan the one or more walls, ceilings and/or floors of the room. The processor **1922** can group and detect scanned points from the scene into planes, which are stored in the AR plane manager **1926**. In some embodiments, the operation **2006** includes detecting planes in the scene using the depth information from the optical camera and/or Lidar camera **1916**. In some embodiments, the detection of planes in the scene of operation **2006** further includes determining, using the information from scan, that planes detected in the scene can be classified as one or more walls, ceilings and/or floors. In some embodiments, the determination of planes detected in the scene further comprises using 3-D camera position information or 6-D camera position and orientation information in association with the depth information.

[0173] The processor **1922** can also operate the camera **1916** to capture the wall, ceiling or floor as an orientation of the user device **1914** is rotate and moved around the room. One or more sensors of the user device **1914** can capture details for determining a camera pose for the camera **1916** as each frame of the wall, ceiling or floor is captured. The camera pose can be saved along with the Lidar information or each could be stored separately with a respective time stamp to match.

[0174] The processor **1922** can create a network mesh covering a scanned scene in operation **2008**. In some embodiments, the network mesh can be a triangular mesh. In some embodiments, the network mesh can be a point cloud. In some embodiments, each surface or point of the mesh is determined based on a distance and/or orientation from an initial reference point. In some embodiments, each surface of the mesh is determined based on a distance and/or orientation from one or more planes in the scene. The network mesh can be determined based on a combination of the Lidar and the camera pose. In certain embodiments, the camera can be used in combination with the camera pose and other sensors on the user device **1914** to create the network mesh. The network mesh can be formed using one or more shapes, such as a triangular mesh.

[0175] The processor **1922** can select a texture for an AR background in operation **2010**. The texture can be based on an image of the scene captured by the camera **1916**. The processor **1916** can analyze the image and determine a texture for the AR background. Different types of textures that can be selected could include color, pattern, etc. The AR background can have one or more textures that can be selected. In certain embodiments, the camera can process an image of a wall in a room and determine a color and pattern for the wall to be used for the texture of the AR background to simulate the color and pattern on the actual wall. Alternatively, the texture of the AR background can be preselected on a user device, selected at the time of mesh generation, or read from a default setting in the memory. While a texture close to the actual background of the scene is preferable in some embodiments, a texture that is different from the background can be chosen in some instances to more clearly distinguish the AR background from objects in the scene.

[0176] The processor **1922** can put the selected texture on the AR background in operation **2012**. Once the texture is selected based on operation **2010**, the processor **1922** can build a mesh for a point cloud or a mesh for a 3D depth image as triangle definitions. The selected texture can be viewed on the display **1920** overlaid as texture on the generated mesh. This mesh is virtually placed in the scene on top of the background currently captured by the camera **1916** of the user device **1914**. A dimension (e.g., depth or distance from the camera) for the background captured from the camera **1916** can be determined and the selected texture can be implemented in front of or behind physical objects in the scene where the background may be not visible. Using the depth value associated with each pixel in the background, the generated mesh with the selected texture can be placed relative (in front of/behind) to the objects in the background. This can be used to selectively hide or show objects in the scene.

[0177] The processor **1922** can replace an actual object in the scene with a corresponding AR shape (i.e., AR object) and at a corresponding location in the scene in operation **2014**. Objects in the scene can be identified in front of the background. Objects in the scene may include, but are not limited to, all types of furnishings and personal property within residence or building such as tables, chairs, sofas, lamps, cabinets, chests, desks, books, bookcases, shelves, beds, stoves, ovens, microwave ovens, dishes, cups, glasses, window curtains, window blinds, pictures, artwork, clocks, vases, rugs, carpets, televisions, radios, electronic equipment (collectively “furnishings”). In some embodiments, the objects in the scene can be identified by evaluating the previously-created mesh for specific predetermined patterns. Such predetermined patterns may include, but are not limited to, mesh contours, mesh contour borders, mesh flatness, mesh depth from background (e.g., distance in front of the wall), mesh orientation from background (e.g., angle to wall). Such predetermined patterns may be stored in the memory **1924** of the user device **1914**. In some embodiments, identified objects in the scene may be classified into one or more categories of like objects. In some embodiments, the method may allow the user to confirm whether the classification of the identified object in the scene is correct. In some embodiments, the user may be able to edit the classification of an identified object. The processor **1922** can determine dimensions, orientation, and position within the scene for each detected object. The processor **1922** can use the dimensions of the object to create a corresponding AR shape and use the orientation and position within the scene for placing the AR shape as an overlay for the object in the display **1920**. The AR shape can correspond to a 3D representation of the object. The processor **1922** can extract and layout an actual image of the object in operation **2016**. The processor **1922** uses the camera **1916** to capture one or more fames and can identify the object in the image frame using the position information to identify the object scene. The portion of the image frame corresponding to the position of the object can be extracted. The extracted portion of the image can be overlaid on the 3D shape as a virtual representation of the object. The virtual representation of the object refers to the 3D shape with the actual image overlaid. The actual image and the 3D shape can be manipulated as the virtual representation of the object.

[0178] For example, in one embodiment the operation **2014** may evaluate the mesh, identify an object in a scene, and then classify the object into a category of like objects, e.g., a framed artwork (i.e., “picture”), by evaluating one or more predetermined patterns in the mesh such as: a) rectangular shape, i.e., mesh face bounded by four straight contour lines; b) flatness, i.e., mesh face at a substantially constant (i.e., within a preselected tolerance) distance above the underlying background plane; c) shallowness, i.e., average distance of face mesh above the underlying background plane is within a preselected tolerance; and/or d) frame presence, i.e., presence of a substantially constant (i.e., within a preselected tolerance) mesh color around the edges of the object. In some embodiments, the method may request user confirmation that the object is correctly classified as a framed artwork. Once the object is identified as framed artwork, the processor **1922** can use the dimensions of the actual object to create a corresponding AR shape with the characteristics of a framed artwork. The processor **1922** may use the orientation and position of the actual object within the scene for initially placing the AR framed artwork as an overlay for the actual framed artwork in the display **1920**. In other words, the AR object, in this case a rectangle, may be initially displayed in front of the actual object in the view shown on the display **1920**. The AR background may be modified to extend over the area coinciding with the actual object in the view shown on the display **1920**. A photo image of the actual object may be applied as a texture to the surface of the AR shape. Thus, the AR shape may initially appear to the user as the same shape and position of the actual object. The processor **1922** can subsequently move the position of the AR framed artwork on the display screen **1920** and the image of the actual framed artwork will be obscured by the AR background so that it is not visible on the display screen. In this manner, the AR framed artwork will appear to the user observing the display screen **1920** to be the actual framed artwork, but the AR framed artwork can be manipulated by the processor **1922**, e.g., to move, to oscillate, to fall and/or to break, despite the fact that the actual framed artwork remains unmoved and intact.

[0179] As shown in the simulation subprocess **2004** of FIG. **20**B, the processor **1922** can select one or more slider values in operation **2018**. The slider values can be controls for a parameter or attribute of the seismic activity. For example, the slider can control an intensity, a time period, etc. Additional parameters or attributes of the seismic activity not controlled by a slider can have values entered prior to the simulation or read from a memory **1918**. The processor **1922** can show a slider or other mechanism for selecting the parameter value or attribute of the seismic activity on the display **1920**. In certain embodiments, the slider can control a current intensity and seismic activity would continuously occur at the selected current intensity until the current intensity is set to zero on the slider. In some embodiments, a touch-screen or physical buttons, voice commands or other inputs can be used for entering parameters or attributes of the seismic activity instead of, or in addition to, the slider.

[0180] The processor **1922** can generate simulated (e.g., AR) seismic activity for the scene in operation **2020**. The AR seismic activity can be shown to the user through the display **1920** of the user device **1914**. The AR seismic activity can cause the portion of the AR background that is currently visible to shift in the frame of view on the display **1920**. The edge of the background can be moved in and out of the frame in the display according to the one or more slider values.

[0181] The processor **1922** can display simulated (i.e., AR) cracks in the AR background shown on the display **1920** based on the slider value in operation **2022**. In some embodiments, the processor **1922** can dynamically generate AR cracks in the AR background during a simulated seismic event, i.e., the cracks may grow in length or width as the seismic event progresses. In some embodiments, the processor **1922** can create a random number of AR cracks with varying paths, widths and/or depths based on the slider value in operation **2022**. In some embodiments, the processor **1922** can generate AR cracks featuring parallax shading to give the cracks the appearance of three-dimensional depth and width based on the point of view. The selected intensity of the simulation can be a factor for the crack generation. The crack generation can also be determined based on an amount of time that has passed and/or a total time of the simulation. For example, additional cracks may be formed as the time of the simulated seismic activity continues. In addition, the dimension of the cracks can change over time and based on the factors of the seismic activity.

[0182] In operation **2024**, the processor **1922** can move or manipulate an AR object (i.e., the virtual representation of the object) shown on the display **1920** during a simulation of a seismic event. In some embodiments, the processor **1922** may determine the type of a movement or a manipulation to be applied to an AR object based on one or more seismic characteristics input by the user or received from the system. In some embodiments, the processor **1922** may determine the type of a movement or a manipulation to be applied to an AR object based on the category of like objects into which the object has been classified. In some embodiments, the processor **1922** may determine the type of a movement or a manipulation to be applied to an AR object based on specified or sensed characteristics of the AR object including, but not limited to the object's mass, size and/or material. In some embodiments, the processor **1922** may determine the type of a movement or a manipulation to be applied to the AR object by applying a combination of the factors described above. Types of movements of the AR object that may be applied by the processor **1922** and shown on the display **1920** include, but are not limited to, translation, rotation, oscillation, falling and rolling. Type of manipulations of the AR object that may be applied by the processor **1922** and shown on the display **1920** include, but are not limited to, cracking, breaking, burning, leaking, sparking and soiling. For example, if an object in the scene is classified as a type of object that may be manipulated with “oscillation,” the processor **1922** can show the AR object swinging back and forth about a hinge point during the seismic event, and the magnitude of the swinging may be selected based on a seismic characteristic set by the slider value and/or on a characteristic of the object, such as its size. The virtual representation of the object can swing in relation to the background of the scene. In addition, if there are multiple AR objects in a scene, each virtual representation may swing independently from other virtual representations. In certain embodiments, a virtual representation of a picture frame can swing on a wall. While the virtual representation is swinging, the actual object is obscured by the wall set as a background in order for the swinging motion to not be disrupted by the actual picture frame.

[0183] In operation **2026**, the processor **1922** can apply secondary manipulation types to the AR objects during a simulation of a seismic event. A secondary manipulation can be any manipulation that is applied only when predetermined conditions are met during a simulated seismic event. For example, if the AR object has a primary manipulation type of “oscillation” during a seismic event, the AR object may have a secondary manipulation type of “fall” if the magnitude of the shaking exceeds a preselected level, or if the shaking duration exceeds a preselected period. Thus, the processor **1922** will initially show the AR object oscillating (i.e., swinging) during a simulated seismic event, but will only show the AR object falling if the preselected conditions are met. If the preselected conditions are not met, the AR object will continue swinging according to the primary manipulation type. In some embodiments, the processor **1922** may show the virtual representation of the object fall to the floor based on a highest slider value in operation **2026**. The virtual representation of the object can realistically drop to the floor using standard physics for a falling motion of the virtual representation. When the virtual representation of the object reaches another object or a ground level, the virtual representation can react with the other object or the ground level. For instance, a force could be applied to the other object based on the virtual representation dropping to the floor. The virtual representation can break or crack by the processor **1922** dividing the virtual representation into two or more part virtual representations. For instance, a picture frame can reach the ground level and cause a crack through the picture frame creating two separate parts of the picture frame. The two parts of the picture frame may operate independently for a remaining time of the simulation.

[0184] The virtual representation and background can be shaken independently. For example, a first picture frame can move independently from a second picture frame and a wall. The processor **1922** can create a set of parent coordinate transforms in an application, such as Unity. The parent transform can be an intermediate coordinate system between the world coordinate and children coordinates. The processor **1922** can set all sub-meshes from all regions that have a first face classification, such as “floor”, to a first transform (also referred to as a parent transform) and all sub-meshes from all regions that have a second classification, such as “wall”, to a second transform (also referred to as a child transform). The processor **1922** can set meshes that belong to a plane selected by a user as well as a plane with render cracks to a third transform. The processor **1922** can apply separate random shaking to each of the parent transforms in order to simulate differently classified virtual objects. Parent meshes ensure that a position of the sub meshes with respect to the parent transform remain constant. However, the position of the parent transform with respect to the world is allowed to change. This allows all meshes assigned to a common classification to move together.

[0185] The shaking and falling of objects, such as picture frames, can be performed by implementing Rigid body mechanics using a physic engine, such as but not limited to Unity Physics Engine, which is a standard game engine functionality provided by Unity. The standard game engine can simulate accelerations, connections, joints, and collisions between virtual objects. The selected wall plane and each picture frame can be assigned rigid body functionalities, which allows the objects to interact with each other in forms such as collisions, forces, constraints, and accelerations. The physics engine can provide functions that allow rigid bodies to relate to each other in a form of constraints. The physics engine can then iteratively solve multi-body simulation to give a position of the rigid bodies given the current position, constraints, and external forces/accelerations. The overall steps for this procedure can include (1) assigning rigid body functionalities to various identified AR objects in the scene, (2) adding constraints, such as joints, links, etc. between the instantiated virtual objects, and (3) adding acceleration or movement to one or more AR objects.

[0186] Although FIGS. **20**A and **20**B illustrate an example method for an earthquake effect visualization process **2000**, various changes may be made to FIGS. **20**A and **20**B. For example, while shown as a series of steps, various steps in FIGS. **20**A and **20**B may overlap, occur in parallel, occur in a different order, or occur any number of times.

[0187] FIGS. **21**A and **21**B illustrate an example method for scene generation process **2002** in accordance with this disclosure. As shown in FIG. **21**A, the processor **1922** can activate AR plane manager application **1926** in operation **2100**. The AR plane manager application **1926** can create, update and remove objects with AR plane components.

[0188] The processor **1922** can scan planes in a scene in operation **2102**. The planes can be converted to AR planes by the AR plane manager application **1926**. The scanning of the planes can be performed by a Lidar sensor and a camera sensor. While the planes are scanned, other sensors on the user device **1914** can detect different information to be used with lidar information and camera information. Details of the AR plane can include position information, orientation information, bound information, etc. The planes can be classified by the AR plane manager application **1926**. Non-limiting examples of classifications of the planes can include wall, floor, ceiling, door, window, none, etc.

[0189] The processor **1922** can select one or more planes to simulate an earthquake in operation **2104**. The plane can be selected based on a current frame captured by the camera, selected by a user, etc. In certain embodiments, the planes can be selected based on a number of objects associated with the plane, between the camera and the plane, etc.

[0190] The processor **1922** can activate an AR mesh manager application **1928** in operation **2106**. The AR mesh manager application **1928** can manage network meshes generated by the user device **1914**. The AR mesh manager application **1928** can create, update, and/or remove virtual objects in response to the environment.

[0191] The processor **1922** can scan for meshes in operation **2108**. The AR mesh manager application **1928** can be used to manage, store in memory, or perform operations on the meshes scanned by the processor. The meshes can be defined and/or categorized by the AR mesh manager application **1928** as belonging to various object classes such as but not limited to wall, floor, door, window, seat, none, etc. Meshes can be associated with objects in the scene in addition to a background. For example, meshes can be defined as part of a wall and part of one or more picture frames on the wall.

[0192] The processor **1922** can activate an AR camera manager application **1930** in operation **2110**. The AR camera manager application **1930** can provide texture information and light estimation information. The AR camera manager application **1930** can also provide camera calibration parameters such as focal length, principle point, etc. The AR camera manager application **1930** can extract a camera image and camera parameters (such as camera pose parameters).

[0193] The processor **1922** can set global variables for the earthquake and also data extracted from the scanning process in operation **2114**. The global variables can be received from a user selection on the display **1920** or input of the user device **1914**. Non-limiting examples of the global variables can include seismic intensity, seismic duration, direction of seismic origin, etc. The global variables can also include variables associated with the selected plane, meshes that are not associated with the plane, meshes that are associated with the plane, virtual representations of object, etc.

[0194] The processor **1922** can extract a color of a selected plane in operation **2116**. In certain embodiments, the user device **914** can receive an input from a user for selecting a color of the selected plane, such as a tap on the display **1920** of the user device **1914**. The AR camera manager application **1930** can provide a camera image and the processor **1922** can determine a location in the camera image corresponding to a location of the tap on the display **1920**. The processor **1922** can determine a color at the location in the camera image to extract. In certain embodiments, the processor **1922** can extract a color based on other factors, such as but not limited to color at user tap position, an average color of the wall, a pattern on the wall, etc.

[0195] The processor **1922** can set a color for a mesh according to the extracted color in operation **2118**. The AR camera manager application **1930** and/or processor **1922** can take the extracted color and apply the color to the mesh corresponding to the background of the scene. In certain embodiments, the AR camera manager application **1930** can apply the extracted color to a wall in a room.

[0196] The processor **1922** can extract object suggestions from the mesh in operation **2120**. The processor **1922** can identify meshes corresponding to one or more objects in the scene. The meshes of different objects in the scene can overlap and the processor **1922** can distinguish different objects using the AR camera manager application **1930**.

[0197] The processor **1922** can extract objects from the mesh belonging to a selected scene in operation **2122**. Meshes that are not assigned to one or more of the planes, backgrounds, walls, etc. can be determined by the processor to be objects. The objects can be individually processed by the processor **1922** for details of the objects, such as dimensions, orientation, position, etc. The extraction of the objects from the mesh is described in greater detail corresponding to the FIGS. **23**A and **23**B.

[0198] The processor **1922** can extract images for an object in a camera frame in operation **2124**. The processor **1922** can extract one or more camera images supplied by the AR camera manager application **1930**. The camera images can be continuously captured based on movement of the AR camera in order to capture the object from different angles. In certain embodiments, the processor **1922** can capture camera images including one or more picture frames.

[0199] The processor **1922** can determine that vertexes of the objects are clearly visible in the camera frame in operation **2126**. Using the AR camera manager application **1930**, the processor **1922** can identify each vertex of an object shown in the camera image. Camera images where the object is only partially visible can be skipped or used for clearly identified vertexes.

[0200] The processor **1922** can extract an image corresponding to the object in operation **2128**. The details of the objects can be used by the processor **1922** to identify an actual object in the camera image and extract a partial image for the actual object. For example, the processor **1922** can identify a picture frame in one or more camera images and make sure that all the corners of the picture frame are clearly visible in each camera image. The extraction of the image corresponding to the objects is described in greater detail corresponding to the FIGS. **24-26**C.

[0201] The processor **1922** can instantiate selected objects in operation **2130**. The processor **1922** uses the AR camera manager application to overlay the actual image of the object over the corresponding meshes associated with the object in the AR scene. For example, an actual image of a picture frame can be overlaid over meshes associated with the picture frame.

[0202] Although FIGS. **21**A and **21**B illustrate an example method for scene generation process **2002**, various changes may be made to FIGS. **21**A and **21**B. For example, while shown as a series of steps, various steps in FIGS. **21**A and **21**B may overlap, occur in parallel, occur in a different order, or occur any number of times.

[0203] FIG. **22** illustrates example meshes for assigning to a plane in accordance with this disclosure. As shown in FIG. **22**, meshes can be assigned to a plane, such as in operation **2112**. A goal of operation **2112** is to separate meshes into belonging to a plane or not belonging to a plane, such as a wall. The AR mesh manager application **1928** can define each mesh, such as a first mesh **2200** and a second mesh **2202**, as a list of vectors, such as a first vertex **2204**, a second vertex **2206**, a third vertex **2208**, and a fourth vertex **2210**, in a 3D space, a triangle defection, a face classification, etc. For example, the vectors in 3D space can be defined as ([0, 0, 0], [1, 0, 0,], [1, 1, 0], [0, 1, 0]). The triangle definition can be defined by the vertices as ([0, 2, 1], [0, 3, 2]), where the first mesh **2200** is formed of the first vertex **2204**, the third vertex **2208**, and the second vertex **2206**, and the second mesh **2202** is formed of the first vertex **2204**, the fourth vertex **2210**, and the third vertex **2208**. The face classifications can be defined as wall or none as shown in FIG. **22**.

[0204] The AR wall plane that was selected during the plane scan process provides information such as plane position and orientation, plane boundaries in a plane space coordinate, and a 3D parametric plane equation [a, b, c, d] such that ax+by+cz+d=0 for all 3D points [x, y, z] that lie on the plane.

[0205] Using this information, the processor **1922** to perform operation **2112** uses a function to separate mesh points from belonging to a room plane and not belonging to the room plane, depending on whether the points satisfy the following conditions. (1) Points can be at most 0.1 m from a plane surface. Therefore, the at most absolute value of distance of a point [x1, y1, z1] to plane [a, b, c, d] is less than 0.1 m. (2) The point, lies within the plane boundaries to check if a point lies inside the plane boundaries. First transform the point from world space to plane space. Let x_W=[x1, x2, x3] be the point in world space and x_P=[xp1, xp2, xp3] be the point in plane space, R_p be the rotation matrix of the plane in world space, and P=[P1, P2, P3] be the position of the plane in world coordinates. Then the transformation shown in equation 1 holds and the inverse transformation is shown in equation 2.

[00001] $x\_W = R\_p * x\_P + P$ (1) $x\_P = \text{Inverse}(R\_p) * (x\_W - P)$ (2)

[0206] (3) The processor **1922** can check whether the point in plane space x_P lies inside the boundaries of the plane. The y coordinate of the point can be dropped in order to obtain a corresponding 2D coordinate of the point in the plane space. i.e. x_P=[xp1, xp2, xp3]==[xp1, xp3] in the 2D plane space, since the normal of the plane is the local y axis. (4) Finally to find out whether the point x_P belongs to the plane or outside the plane, the processor **1922** can compute the Winding Number for the points. If the winding number of the point does not equal to 0, then the point lies inside the plane boundaries. If the winding number of the point equals 0, the point lies outside the plane boundaries.

[0207] Although FIG. **22** illustrates an example meshes for assigning to a plane, various changes may be made to FIG. **22**. For example, various components in FIG. **22** may be combined, further subdivided, replicated, omitted, or rearranged and additional components may be added according to particular needs.

[0208] FIGS. **23**A and **23**B illustrate an example object extraction **2300** from mesh in accordance with this disclosure. As shown in FIG. **23**A, an object, such as a picture frame, can be extracted from a mesh. From operation **2112**, the meshes can be categorized as a wall mesh **2302** or a none mesh **2304** at this point. Using the AR mesh manager **1928**, the processor **1922** can get the classification for each triangle in the mesh. The processor **1922** can identify the triangles that are classified as “none”. The bounds of the disjointed meshes **2306** that have a classification of “none” can be used as initial bound cuboid **2308** as picture frame proposals.

[0209] The processor **1922** can use a modified version of the union find algorithm with path optimization to determine the bounds **2306** of the disjointed meshes. The union find algorithm was originally designed to solve Kruskal's minimum spanning tree algorithm. The main goal of the union find algorithm is to have an efficient method of extracting and storing disjoint sets of nodes in a tree. The processor **1922** can find the disjoined meshes **2306** of vertices inside a mesh that belongs to the class “none”. As a non-limiting example, two picture frames exist inside of a single Mesh definition and need to be separated into two sub-meshes. The processor **1922** can use union operation to group two nodes into a single group and a find operation to find a parent of the group to which a node belongs to. This method can be adapted by assigning nodes to each vertex **2204-2210** in the mesh as a node and use the triangle definitions as union operations as shown in FIG. **23**B.

[0210] As shown in FIG. **23**B, each vertex **2204-2210** is assigned individual parent meshes. The goal is to set the parents of those vertices that are joined by a line to a common parent. According to the triangle definition [0, 3, 2] results in lines joining the indices [0, 3], [3, 2], [0, 2], which leads to join operations [0, 3], [3, 2], [0, 2]. The first step uses union operation [0, 2] to assign the third vertex **2208** as a child of the first vertex **2204**. The second step uses operation [3, 2] to assign the second vertex **2206** as a child of the third vertex **2208**. The third step uses operation [0, 3] but both have already been assigned to the group making this operation redundant. This operation is repeated for each triangle with a face classification as “none”. The maximum bounding cube **2308** can be computed by extracting the maximum span of all the vertices of each sub-mesh. The bounding cube **2308** is then displayed to the user as a prospective object, such as a prospective picture frame.

[0211] Although FIGS. **23**A and **23**B illustrates an example object extraction **2300** from mesh, various changes may be made to FIGS. **23**A and **23**B. For example, various components in FIGS. **23**A and **23**B may be combined, further subdivided, replicated, omitted, or rearranged and additional components may be added according to particular needs.

[0212] FIG. **24** illustrates an example method for a crack creation algorithm **2400** in accordance with this disclosure. As shown in FIG. **24**, a crack creation algorithm can utilize shader processes that rely on modern mobile GPUs for efficient parallel processing, allowing real-time display of many previously expensive graphics algorithms. The crack creation algorithm used here is performed in two main passes. The first pass utilizes Voronoi noise functions to create a crack-like grayscale depth map. This pass solely requires standardized UV coordinates as an input (as well as custom variables for modifying noise density, width, etc). This shader pass is saved to memory via a texture buffer (known in Unity as a custom render texture [CRT]) to be sent to the second shader pass. The second pass utilizes a technique known as parallax occlusion mapping (POM) to cheaply create a sense of 3D depth. As inputs, this pass requires mesh UVs, mesh normal vectors, camera view direction, and a texture (the CRT from the initial pass), as well as depth amount custom parameter. POM requires a depth texture for sampling and thus it is necessary to create a texture in the first pass that is then passed to the POM algorithm in a second pass.

[0213] The crack creation algorithm **2400** can utilize UV mapping **2402** to project the shader algorithm **2404** onto the chosen mesh. The UV mapping algorithm **2402** is described in greater detail in FIGS. **25**A and **25**B. The Voronoi noise algorithm **2404** is described in greater detail in FIGS. **26**A-**26**C.

[0214] The crack creation algorithm **2400** can utilize a mesh **2408** and a camera image **2410** to form pseudo 3D cracks on a plane. The crack creation algorithm **2400** can obtain mesh information such as UV coordinates and normal (x, y, z) vectors **2412** from the chosen mesh **2408**. The camera image **2410** can include other details, such as camera pose information, that can be used to determine a view direction **2414** that is applied in the parallax occlusion mapping (POM) pass.

[0215] Voronoi noise is a noise function that utilizes distance functions between tiled points to obtain consistent cell like boundaries and spaces. To create a randomized domain of the Voronoi noise, the Voronoi noise pass algorithm **2405** can choose a random seed that connects to multiple randomizers. For example, the random seed can be chosen based on a time in the application. The randomizers can take a standardized UV input (linear gradient from 0-1 on x and y axis) and offset/scale the standard UV coordinates (0-1) randomly in order to generate different variations of noise for displacement. To obtain more vertical or horizontal cracks, the scaling of the UV coordinates can be adjusted separately. These UV values are plugged into two Voronoi noise generation subshaders, a subshader used for the U values **2504** (X-axis) and a subshader used for the V values **2506** (Y-axis). The two subshaders can be centered around 0 (−0.5 to 0.5) and combined into a single image consisting of U displacement on a red channel and V displacement on a green channel. The processor **1922** can use a multiply value to scale a displacement and/or distortion amount to be applied to the main Voronoi generation. The displacement/distortion amount can be added to an initial UV coordinate of the mesh geometry in a process known as domain warping. This warped UV space is then plugged into a final Voronoi subshader as the UV input. The processor **1922** can clamp the output using a smoothstep function to create sharp cell-like edges. This is saved into the memory via a custom render texture feature from Unity.

[0216] The custom render texture algorithm **2418** can allow an output of a shader to be saved into a texture and passed to various locations with Unity and allow for multiple passes. Within the crack plane creation script **2400**, the Voronoi depth shader can be called with desired parameters to render into a texture via the custom render texture algorithm **2418**.

[0217] The texture from the custom render texture algorithm **2418** can be used as an input to the parallax occlusion mapping (POM) pass algorithm **2406**. The POM pass algorithm **2406** can apply shader to a mesh material **2416** as a final output to be shown with the display.

[0218] Although FIG. **24** illustrates an example method for scene generation process **2002**, various changes may be made to FIG. **24**. For example, while shown as a series of steps, various steps in FIG. **24** may overlap, occur in parallel, occur in a different order, or occur any number of times.

[0219] FIGS. **25**A and **25**B illustrate an example UV mapping algorithm **2402** for the crack creation algorithm **2400** in accordance with this disclosure. As shown in FIGS. **25**A and **25**B, UV checker maps **2500** and **2501** are examples for showing how to fit a texture onto a rectangular surface while keeping an aspect ratio. In certain embodiments, the UV checker map can be replaced by the crack shader. In certain embodiments, crack width and depth can be adjusted based off a scaling amount of the UVs to cheaply fit the shader onto various shaped walls. UV mapping can be used to project a procedurally generated texture onto the selected wall plane geometry from a capture process. The captured geometry can designate UV coordinates to a specific location on the mesh where 2D textures can be mapped. In certain embodiments, the values for the 2D texture can be between 0.0 and 1.0.

[0220] The processor **1922** can hold the UV blocks **2502** as a steady size. Therefore, a number of UV blocks **2502** is proportional to a size of the wall and, as the wall size increases, more UV blocks **2502** are utilized. UV blocks **2502** can be normalized between 0 and 1 in order to keep a constant aspect ratio. To normalize, an absolute value of the minimum value for the U value **2504** and V value **2506** to the actual values of the U value **2504** and the V value **2506** for each UV block **2502**. The actual values of the U value **2504** and the V value **2506** for each UV block **2502** can be divided by a maximum value between the U values **2504** and the V values **2506**. For example, the maximum value for the U value **2504** is 3.0 and for the V value **2506** is 2.0, which means the maximum value used to divide with is 3.0 resulting in a normalized UV table **2508** of normalized UV blocks **2510**.

[0221] Although FIGS. **25**A and **25**B illustrate an example UV mapping for the crack creation algorithm, various changes may be made to FIGS. **25**A and **25**B. For example, various components in FIGS. **25**A and **25**B may be combined, further subdivided, replicated, omitted, or rearranged and additional components may be added according to particular needs.

[0222] FIGS. **26**A-**26**C illustrate an example Voronoi noise pass algorithm **2405** for the crack creation algorithm **2400** in accordance with this disclosure. While the Voronoi noise pass algorithm **2405** is described being performed by processor **1922**, the Voronoi noise pass algorithm **2405** can also be performed as a shader pass using a parallel computation graphic processing unit. In particular, FIG. **26**A illustrates an individual cell distance function **2600**, FIG. **26**B illustrates a neighbor cell distance function **2602**, and FIG. **26**C illustrates a Voronoi block **2604**. As shown in FIGS. **26**A-**26**C, a Voronoi noise pass algorithm **2405** can be used to define and delineate proximal regions around UV blocks **2502** using the boundaries of the UV blocks **2502**. The processor **1922** can subdivide the normalized UV block **2510** from the UV table **2508** to get smaller square cells **2606**. The processor **1922** can randomly generate a point **2608** in the cells **2606**. A distance can be calculated for each pixel in the cell **2606** from the point **2608** in the respective cell **2606**.

[0223] The Voronoi noise pass algorithm **2405** can generate a usable seamless distance function by searching neighboring cells **2610**. The processor **1922** can calculate a distance from each pixel to a nearest point either in a respective cell or a neighboring cell. This distance calculation is performed to determine a minimum distance to a point from the current pixel across all searched cells. The neighbor cell distance function **2602** is shaded to reflect this distance calculation for each pixel to the closest point in any cell, as shown in FIG. **26**B.

[0224] As shown in FIG. **26**C, the Voronoi block **2604** has the borders from the cells **2606** redrawn to show Voronoi cells **2612**. Instead of a search of one cell **2606** around a current cell, a more accurate two cell search can be performed. When points in neighbor cells are close together, a one cell radius can lead to distortions near borders. To create an edge effect on the Voronoi cells **2612**, a second closest point is tracked along with the closest point to calculate the distance to cell borders. A distance function of the second closest point can be assigned a variable F2 and a distance function of the closest point can be assigned a variable F1. Using F1 and F2, the processor **1922** can determine pseudo-border on the Voronoi cells **2612** with help of a smoothstep function and/or clamping function.

[0225] Although FIGS. **26**A-**26**C illustrate an example Voronoi noise **2600** for the crack creation algorithm, various changes may be made to FIGS. **26**A-**26**C. For example, various components in FIGS. **26**A-**26**C may be combined, further subdivided, replicated, omitted, or rearranged and additional components may be added according to particular needs.

[0226] FIG. **27** illustrates an example AR earthquake visualization and assessment **2700** in accordance with this disclosure. In particular, FIG. **27** illustrates the user's point of view as the user holds a mobile device directed towards a portion of the room (the user's hand is not shown). The system is shown generating an AR earthquake scene on the display screen based on the actual room scene visible behind the mobile device. As shown in FIG. **27**, The AR earthquake visualization and assessment **2700** can include a scene **2702** and a user device **2704**. The scene **2702** includes a wall **2706**, a floor **2708**, a window **2710**, a first picture **2712**, a second picture **2714**, a chair **2716**, and a rug **2718**. However, the scene **2702** can include any other type of object and any number of a specific object.

[0227] The user device **2704** can include a display **2720**, one or more sensors **2722** (e.g., optical camera, time of flight camera, etc.), and one or more physical inputs **2724** (e.g., buttons, switches, etc.). The user device **2704** can use a scan area **2723** one or more sensors **2722** to scan the scene **2702** and detect the background (e.g., wall **2706** and floor **2708**) and objects (a window **2710**, a first picture **2712**, a second picture **2714**, a chair **2716**, and a rug **2718**). The user device **2704** can create virtual or AR representations of the scene **2702** and present representations on the display **2720**. For example, the display **2720** shows an AR wall **2726**, an AR floor **2728**, an AR window **2730**, an AR first picture **2732**, an AR second picture **2734**, an AR chair **2736**, and an AR rug **2738** corresponding to the respective backgrounds **2706** and **2708** and objects **2710-2718**.

[0228] The AR backgrounds and AR objects can have characteristics individually determined applied during and/or after the scanning process. For example, a weight, material, roughness factor, etc. can be estimated for an AR object. The characteristics can be used for each manipulation of the AR background and the AR objects. In certain embodiments, a value can be assigned to each of the AR backgrounds and the AR objects. The value can be based on a purchase value, an actual value, an input value, a replacement value, etc. The values can be stored in a table, such as inventory table **2800** shown below in FIG. **28**, in the memory of the user device **2704**, which can be viewed separately or in combination with the AR environment. The values can also be shown with AR objects as each AR object is selected or as the value is determined. The values that are shown with the AR objects can disappear after a period of time or approval by the user. The values can be adjusted by the user.

[0229] Values for damage can be determined based which of the following AR effects are applied to a respective AR background or AR object. The values for damage can be based on a repair value, a replacement value, etc. The values for damage can be calculated based on an estimated amount of damage and for an amount of damage exceeding a threshold, an AR object can be considered fully damaged or destroyed. The values for damage can be saved in the table in the memory of the user device **2704**. The damage values can be shown on the respective AR background or AR object that receives the damage. The damage values can disappear after a period of time or selection from a user. The damage values can include a total damage for the seismic activity applied to AR representation of the scene **2702**.

[0230] As shown in FIG. **27**, the AR background and AR objects can be manipulated on the display **2720** while remaining unaffected in the scene **2702**. As the user device **2704** is moved around the room, the display **2704** updates a current status of the objects. The one or more sensors **2722** can capture the actual room and the display can show the room in a current state of seismic activity. The display **2720** can also display controls (such as a seismic attribute control **2740** and a start button **2742**). The seismic attribute control **2740** includes an input **2744** that can be adjusted. The seismic attribute control **2740** can control the seismic attribute, such as intensity, duration, etc., by controlling the input **2744**. In the illustrated example, the input **2744** can be a slide mechanism, value entry, knob, etc. The input **2744** can be controlled on a touchable display or using the one or more physical inputs **2724**. The start button **2742** initiates a seismic event on the display based on the input **2744** provided on the seismic attribute control **2740**.

[0231] The display **2720** can apply a vibration or shaking effect **2746** to an AR background or an AR object, such as the AR wall **2726** and the AR floor **2728**. The shaking effect **2746** can be applied equally to the AR backgrounds or can be applied stronger or weaker to the specific AR backgrounds. The shaking effect **2746** applied to the AR background can be additionally applied to each of the AR objects in a manner of equal characteristics or reduced effectiveness characteristics. The reduced effectiveness characteristics can include lower shaking intensity transferred from the background, lower resistance (increased duration of shaking), etc. An amount of the vibration for the AR background or AR object can be controlled based on the input **2744** to the seismic attribute control **2740**.
[0232] The display **2720** can apply a cracking effect **2748** on an AR background or an AR object, such as the AR wall **2726**. The cracking effect **2748** can cause a 3D crack to form in the AR background. The 3D crack can be viewed from different angles to show a depth to the 3D crack. The cracking effect **2748** can include a 3D crack that extends from a first AR background to a second AR background. For example, the cracking effect **2748** can begin on the AR wall **2726** and extend to an AR ceiling, an AR second wall, and/or the AR floor **2728**. A 3D crack on an AR background, such as the AR wall **2726**, can extend behind and hidden an AR object to continue extending past the AR object in view on the display **2720**. A size and growth rate for the 3D crack on the AR background or AR object can be controlled based on the input **2744** to the seismic attribute control **2740**.
[0233] The display **2720** can apply a swinging effect **2750** to an AR object, such as the AR first picture **2732**. The swinging effect **2750** can cause the AR object to shake similarly to the AR background contacting the AR object due to a fixed pendulum that the swinging occurs about. The swinging effect **2750** on the AR object can include a translation component equal to the vibration or shaking effect **2746** of the AR background. A swing rate for the swinging of AR object can be controlled based on the input **2744** to the seismic attribute control **2740**.
[0234] The display **2720** can apply a falling effect **2752** to an AR object, such as the AR second picture **2734**. The falling effect **2752** can move the AR object separately from the AR background and other AR objects. The falling effect **2752** can be based on gravity for controlling a falling rate and/or a falling speed. The falling effect **2752** can use an estimated or determined weight of the AR object to determine a force that could be applied to other AR objects and/or AR backgrounds. A timing for the falling of AR object can be controlled based on the input **2744** to the seismic attribute control **2740**.
[0235] In certain embodiments, a secondary effect can be applied to an AR object, such as the AR second picture **2734**. For example, a falling effect on an AR object causes the AR object to forcefully contact an AR background, such as the AR floor **2728**, or another AR object, such as the AR chair **2736**. A secondary effect of the falling effect could be a breaking effect **2754**. The display can apply the breaking effect **2754** to the AR object, such as the AR second picture **2734**. Other examples of secondary effects can include, a bouncing effect, a ricochet effect, a merge effect, striking effect, etc.
[0236] The breaking effect **2754** can cause the AR object to split into multiple AR partial objects. Each of the multiple AR partial objects can be smaller than the AR object before the split. In certain embodiments, the multiple AR partial objects can collectively be approximately equal to the broken AR object. An amount of AR partial objects can be based on an original size of an AR object, a force applied to the AR object during simulation from one or more other AR effects, etc.
[0237] A shatter effect **2756** can cause an AR object to have multiple AR partial objects splits from a main AR object, such as AR window **2730** or an AR mirror. The shatter effect **2756** can be a primary effect from the seismic damage or a secondary effect from another AR object forcefully interacting with the original AR object. For example, the AR window **2730** can shatter due directly to the vibrations from a seismic event or from an AR object, such as the AR second painting **2734** or generation of a random AR object, such as an AR branch of a tree, an AR power line, etc., falling

through the AR object, such as AR window **2730**.

[0238] Although FIG. **27** illustrates an example AR earthquake visualization and assessment **2700**, various changes may be made to FIG. **27**. For example, various components in FIG. **27** may be combined, further subdivided, replicated, omitted, or rearranged and additional components may be added according to particular needs.

[0239] FIGS. **28**A and **28**B illustrate an example inventory tables for objects and backgrounds identified in a scene **2702** in accordance with this disclosure. In particular, FIG. **28**A illustrates an example object inventory table **2800** and FIG. **28**B illustrates an example background inventory table **2801**. As shown in FIG. **28**A, an object inventory table **2800** can be populated with AR objects identified in a scene scanning process above. The inventory table can include an object number **2802**, an object category **2804**, an initial value **2806**, a custom value **2808**, an assigned value **2810**, a damage factor **2812**, and a damage value **2814**. In certain embodiments, a new table can be generated for each newly identified scenes. When a scene identifies a previously scanned scene, the original table corresponding to the previously scanned scene can be amended or added. For instance, the objects added to the previously scanned scene can be added to additional rows of the table. Objects that no longer appear in the previously scanned scene can be removed from the table, a note can be added to the object row in the table, or otherwise marked as no longer in the previously scanned scene.

[0240] The object number **2802** can be an ordered number assigned for an object. The object number **2802** can be a unique number for the scanned scene. In certain embodiments, objects in different scenes are given unique object numbers **2802** for any scene that the user device has captured. In certain embodiments, object numbers **2802** can be unique for a specific location. In certain embodiments, each location can have object numbers **2802** assigned as series. For example, the object number **2802** can be four digits and the first two digits are assigned a grouping of scenes and the second two digits are assigned in order for each scene.

[0241] The object category **2804** can be a general narrative for the identified object. In order to determine an object category **2804**, the user device can compare the captured image to a database of standard objects. In certain embodiments, the identification of the object can be performed by a server sent an image, video, or point cloud of the identified object. The object can be identified by any standard object identification method. The object category **2804** can be identified by the user primarily or secondarily after one or more of the user device and a remote server cannot match a description to the object. The object category **2804** can also provide a fragility characteristic of the object. For example, a vase on a table could provide a description of warning regarding possibility of destruction, damage, secondary damage, etc.

[0242] The initial value **2806** can be determined for each of the scanned objects. The processor **1922** can determine an initial value **2806** based on the object category **2804**. The initial value **2806** can be a value of an object determined by the user device or the remote server. The initial value **2806** could be a minimum value, a medium value, a mean value, a maximum value, etc. The initial value **2806** could be a wholesale value, a purchase value, a market value, a replacement value, etc.

[0243] The custom value **2808** can be input by a user. A prompt could be provided to the user after an AR object is selected for generation on the display. The user could input the custom value **2808** using the physical buttons. The user could provide the custom value **2808** by scanning a barcode or other visual code for identifying an object. In certain embodiments, the custom value **2808** can be entered on another remote user device.

[0244] The assigned value **2810** is the value used for totaling a total value of items in a scene. The assigned value **2810** can be determined based on whether a value is entered in the custom value **2808**. For example, the assigned value **2810** can be the custom value **2808** when a value is entered for the input value **2808** or the custom value **2808** is greater than 0. The assigned value **2810** can be the initial value **2806** when the custom value **2808** is not entered or the custom value **2808** is approximately equal to 0 or an insignificant value compared to the initial value **2806**.

[0245] The damage factor **2812** can be a factor related to damage of the object. The damage factor **1812** can be determined based on a threshold level earthquake, a historic level earthquake (for a region), a maximum earthquake, or an earthquake controlled on the display by the user. The damage factor **1812** can be a factor related to an object being considered totally destroyed or a factor of damage where an object is considered destroyed. The damage factor **1812** could be a factor related to a chance that an object is destroyed in an average or low damage producing earthquake. In certain embodiments, a scene **2702** can be scanned following an actual earthquake and the damage factor **2812** can reflect an amount of actual damage of the identified objects. The damage factor can be determined based on the seismic effects applied to the specified object.

[0246] The damage value **2814** is a value for the amount of damage cause by the seismic event. The damage value **2814** can be determined based on the seismic effects applied to the specified object. The damage value **2814** can be determined by multiplying the assigned value **2810** by the damage factor **2812**.

[0247] As shown in FIG. **28**B, a background inventory table **2801** can be populated with AR backgrounds identified in a scene scanning process above. The inventory table can include a background number **2816**, a background category **2818**, an area **2820** of the background, an initial unit value **2822**, a custom unit value **2824**, an assigned value **2826**, a damage factor **2828**, and a damage value **2830**. In certain embodiments, a new table can be generated for each newly identified scenes. When a scene identifies a previously scanned scene, the original table corresponding to the previously scanned scene can be amended or added.

[0248] The background number **2816** can be an ordered number assigned for a background. The background number **2816** can be a unique number for the scanned scene. In certain embodiments, background in different scenes are given unique background numbers **2816** for any scene that the user device has captured. In certain embodiments, background number **2816** can be unique for a specific location. In certain embodiments, each location can have background number **2816** assigned as series. For example, the background number **2816** can be four digits and the first two digits are assigned a grouping of scenes and the second two digits are assigned in order for each scene. The background numbers **2816** can also be unique from the object numbers **2802**.

[0249] The background category **2818** can be a general narrative for the identified background. In order to determine a background category **2818**, the user device can compare the captured image to a database of standard backgrounds. In certain embodiments, the identification of the background can be performed by a server sent an image, video, or point cloud of the identified background. The background can be identified by any standard background identification method. The background category **2818** can be identified by the user primarily or secondarily after one or more of the user device and a remote server cannot match a description to the background.

[0250] The background area **2820** is an area of the background in the scene. The processor **1922** can determine the background area **2820** during the scanning process. A time of flight camera can capture edges of a background, such as a wall, and calculate a total size for the background area **2820**. The background area **2818** can be any standard measurements units.

[0251] The initial unit value **2822** can be determined for each of the scanned backgrounds. The processor **1922** can determine an initial unit value **2822** based on the background category **2818**. The initial unit value **2822** can be a value per measurement unit of the background area **2820**. The initial unit value **2822** can be a value of a background determined by the user device or the remote server. The initial unit value **2822** could be a minimum value, a medium value, a mean value, a maximum value, etc.

[0252] The custom unit value **2824** can be input by a user. A prompt could be provided to the user after an AR background is selected for generation on the display. The user could input the custom unit value **2824** using the physical buttons. The user could provide the custom unit value **2824** by scanning a barcode or other visual code for identifying a background. In certain embodiments, the custom unit value **2824** can be entered on another remote user device.

[0253] The assigned value **2826** is the value used for totaling a total value of items in a scene. For the background assigned value **2826**, the value is a total of the initial unit value **2822** or custom unit value **2824** multiplied by the background area **2820**. The assigned value **2826** can be determined based on whether a value is entered in the custom unit value **2824**. For example, the assigned value **2826** can be the custom value **2824** when a value is entered for the custom value **2824** or the custom value **2824** is greater than 0. The assigned value **2810** can use the initial unit value **2822** when the custom unit value **2808** is not entered or the custom unit value **2824** is approximately equal to 0 or an insignificant value compared to the initial unit value **2822**.
[0254] The damage factor **2812** can be a factor related to damage of the background. The damage factor **1812** can be determined based on a threshold level earthquake, a historic level earthquake (for a region), a maximum earthquake, or an earthquake controlled on the display by the user. The damage factor **1812** can be a factor related to an object being considered totally destroyed or a factor of damage where an object is considered destroyed. The damage factor **1812** could be a factor related to a chance that an object is destroyed in an average or low damage producing earthquake. In certain embodiments, a scene **2702** can be scanned following an actual earthquake and the damage factor **2812** can reflect an amount of actual damage of the identified backgrounds. The damage factor can be determined based on the seismic effects applied to the specified background.
[0255] The damage value **2814** is a value for the amount of damage cause by the seismic event. The damage value **2814** can be determined based on the seismic effects applied to the specified background. The damage value **2814** can be determined by multiple the assigned value **2810** by the damage factor **2812**.
[0256] Although FIGS. **28**A and **28**B illustrate example inventory tables for objects and backgrounds identified in a scene **2702**, various changes may be made to FIGS. **28**A and **28**B. For example, various components in FIGS. **28**A and **28**B may be combined, further subdivided, replicated, omitted, or rearranged and additional components may be added according to particular needs.
[0257] FIG. **29** illustrates an example method for an AR earthquake visualization and assessment process **2900** in accordance with this disclosure. As shown in FIG. **29**, the processor **1922** can scan a scene **2702** using an optical recognition method in operation **2902**. In certain embodiments, the scene **2702** can be an interior of a room and one or more walls **2706** of the room can be scanned. The user device **1914** can further include a time-of-flight camera **1916** or Lidar sensor, which the processor **1922** can operate to scan the one or more walls **2706**, ceilings and/or floors **2708** of the room. The processor **1922** can determine a distance to different points on the wall **2706**, ceiling and/or floor **2708** using Lidar. In some embodiments, the operation **2902** includes detecting planes in the scene **2702** using the depth information from the optical camera and/or Lidar camera **1916**. In some embodiments, the detecting planes in the scene **2702** of operation **2902** further includes determining, using the depth information from scan, that planes detected in the scene **2702** are one or more walls **2706**, ceilings and/or floors **2708** present in the scene **2702**. In some embodiments, the determining that planes detected in the scene **2702** are one or more walls **2706**, ceilings and/or floors **2708** present in the scene **2702** further comprises using 3-D camera position information or 6-D camera position and orientation information in association with the depth information.
[0258] The processor **1922** can also operate the camera **1916** to capture the wall **2706**, ceiling or floor **2708** as an orientation of the user device **1914** is rotate and moved around the room. One or more sensors of the user device **1914** can capture details for determining a camera pose for the camera **1916** as each frame of the wall **2706**, ceiling or floor **2708** is captured. The camera pose can be saved along with the Lidar information or each could be stored separately with a respective time stamp to match.
[0259] The processor **1922** can identify a background and objects in a scene **2702** in operation **2904**. The processor **1922** can scan the scene **2702** for meshes. The meshes can be defined and/or

categorized by the processor **1922**. The meshes can also be associated with a specified plane or specified planes by the processor **1922**. Meshes can be associated with objects in the scene in addition to a background. For example, meshes can be defined as part of a wall and part of one or more picture frames on the wall. In order to identify backgrounds, the processor **1922** can assign one or more meshes from a region to the plane. The processor **1922** can use the extracted camera image to determine a plane for each of the meshes. The meshes can be classified and associated with a specified plane by the processor **1922**.

[0260] The processor **1922** can extract object suggestions from the mesh. The processor **1922** can identify meshes corresponding to one or more objects in the scene **2702**. The meshes of different objects in the scene can overlap and the processor **1922** can distinguish different objects. The processor **1922** can extract objects from the mesh belonging to a selected scene. Meshes that are not assigned to one or more of the planes, backgrounds, walls, etc. can be determined by the processor to be objects. The objects can be individually processed by the processor **1922** for details of the objects, such as dimensions, orientation, position, etc.

[0261] The processor **1922** can count the objects in operation **2906**. Counting the objects can include taking an inventory of the objects in the scene **2702** and placing the information in an inventory table **2800**. The objects can be assigned an object number **2802** that can be used to relate a row on the table to an AR object by the processor **1922**. A description **2804** of the objects can be determined while counting. The description **2804** of the objects can allow a user to quickly and easily determine which row corresponds to a specific AR object or object in a scene. The description **2804** can include a name of the object, a position of the object, characteristics of the object, etc. The counting of the objects can also include an accounting of the objects and an estimated value **2806** or other value can be determined.

[0262] The processor **1922** can create one or more AR backgrounds and one or more AR objects from the background and objects identified in the scene **2702** in operation **2908**. The processor **1922** can extract a color of a selected plane. In certain embodiments, the user device **914** can receive an input from a user for selecting a color of the selected plane, such as a tap on the display **1920** of the user device **1914**. The camera can provide a camera image and the processor **1922** can determine a location in the camera image corresponding to a location of the tap on the display **1920**. The processor **1922** can determine a color at the location in the camera image to extract. In certain embodiments, the processor **1922** can extract a color based on other factors, such as amount of the color in the scene, an average color of the wall, etc. The processor **1922** can set a color for a mesh according to the extracted color. The processor **1922** can take the extracted color and apply the color to the mesh corresponding to the background of the scene. In certain embodiments, the processor **1922** can apply the extracted color to a wall, a floor, and/or a ceiling in a room.

[0263] The processor **1922** can extract images for an object in a camera frame. The processor **1922** can extract one or more camera images supplied by the AR camera. The camera images can be continuously captured based on movement of the AR camera in order to capture the object from different angles. In certain embodiments, the processor **1922** can capture camera images including one or more picture frames. The processor **1922** can determine that vertexes of the objects are clearly visible in the camera frame in operation **2126**. Using the AR camera, the processor **1922** can identify each vertex of an object shown in the camera image. Camera images where the object is only partially in the camera image can be skipped or used for clearly identified vertexes. The processor **1922** can extract an image corresponding to the object in operation **2128**. The details of the objects can be used by the processor **1922** to identify an actual object in the camera image and extract a partial image for the actual object. For example, the processor **1922** can identify a picture frame in one or more camera images and make sure that all the corners of the picture frame are clearly visible in each camera image. The processor **1922** can instantiate selected objects in operation **2130**. The processor **1922** can overlay the actual image of the object over the corresponding meshes associated with the object in the AR scene. For example, an actual image of a picture frame can be overlaid over meshes associated with the picture frame.

[0264] The processor **1922** can receive seismic characteristics in operation **2910**. The seismic characteristics can be received from a user, read from a memory of the user device, received from another user device, received from a remote server. For example, seismic characteristics or seismic algorithm can be stored in a memory of the user device. The user can be presented an input, such as a slider, to control a seismic characteristic, such as a seismic intensity. The slider values can be controls for a parameter or attribute of the seismic activity. For example, the slider can control an intensity, a time period, etc. Additional parameters or attributes of the seismic activity not controlled by a slider can have values entered prior to the simulation or read from a memory **1918**. The processor **1922** can show a slider or other mechanism for selecting the parameter value or attribute of the seismic activity on the display **1920**. In certain embodiments, the slider can control a current intensity and seismic activity would continuously occur at the selected current intensity until the current intensity is set to zero on the slider. In some embodiments, a touch-screen or physical buttons, voice commands or other inputs can be used for entering parameters or attributes of the seismic activity instead of, or in addition to, the slider.

[0265] The processor **1922** can display seismic effects based on the seismic characteristics in operation **2912**. The processor **1922** can generate simulated (e.g., AR) seismic activity for the scene in operation **2020**. The AR seismic activity can be shown to the user through the display **1920** of the user device **1914**. The AR seismic activity can cause the portion of the AR background that is currently visible to shift in the frame of view on the display **1920**. The edge of the background can be moved in and out of the frame in the display according to the one or more slider values.

[0266] The processor **1922** can display simulated (i.e., AR) cracks in the AR background shown on the display **1920** based on the slider value in operation **2022**. In some embodiments, the processor **1922** can dynamically generate AR cracks in the AR background during a simulated seismic event, i.e., the cracks may grow in length or width as the seismic event progresses. In some embodiments, the processor **1922** can create a random number of AR cracks with varying paths, widths and/or depths based on the slider value in operation **2022**. In some embodiments, the processor **1922** can generate AR cracks featuring parallax shading to give the cracks the appearance of three-dimensional depth and width based on the point of view. The selected intensity of the of the simulation can be a factor for the crack generation. The crack generation can also be determined based on an amount of time that has passed and/or a total time of the simulation. For example, additional cracks may be formed as the time of the simulated seismic activity continues. In addition, the dimension of the cracks can change over time and based on the factors of the seismic activity.

[0267] In operation **2024**, the processor **1922** can move or manipulate an AR object (i.e., the virtual representation of the object) shown on the display **1920** during a simulation of a seismic event. In some embodiments, the processor **1922** may determine the type of a movement or a manipulation to be applied to an AR object based on one or more seismic characteristics input by the user or received from the system. In some embodiments, the processor **1922** may determine the type of a movement or a manipulation to be applied to an AR object based on the category of like objects into which the object has been classified. In some embodiments, the processor **1922** may determine the type of a movement or a manipulation to be applied to an AR object based on specified or sensed characteristics of the AR object including, but not limited to the object's mass, size and/or material. In some embodiments, the processor **1922** may determine the type of a movement or a manipulation to be applied to the AR object by applying a combination of the factors described above. Types of movements of the AR object that may be applied by the processor **1922** and shown on the display **1920** include, but are not limited to, translation, rotation, oscillation, falling and rolling. Type of manipulations of the AR object that may be applied by the processor **1922** and shown on the display **1920** include, but are not limited to, cracking, breaking, burning, leaking, sparking and soiling. For example, if an object in the scene is classified as a type of object that may be manipulated with "oscillation," the processor **1922** can show the AR object swinging back and forth about a hinge point during the seismic event, and the magnitude of the swinging may be selected based on a seismic characteristic set by the slider value and/or on a characteristic of the object, such as its size. The virtual representation of the object can swing in relation to the background of the scene. In addition, if there are multiple AR objects in a scene, each virtual representation may swing independently from other virtual representations. In certain embodiments, a virtual representation of a picture frame can swing on a wall. While the virtual representation is swinging, the actual object is obscured by the wall set as a background in order for the swinging motion to not be disrupted by the actual picture frame.

[0268] In operation **2026**, the processor **1922** can apply secondary manipulation types to the AR objects during a simulation of a seismic event. A secondary manipulation can be any manipulation that is applied only when predetermined conditions are met during a simulated seismic event. For example, if the AR object has a primary manipulation type of "oscillation" during a seismic event, the AR object may have a secondary manipulation type of "fall" if the magnitude of the shaking exceeds a preselected level, or if the shaking duration exceeds a preselected period. Thus, the processor **1922** will initially show the AR object oscillating (i.e., swinging) during a simulated seismic event, but will only show the AR object falling if the preselected conditions are met. If the preselected conditions are not met, the AR object will continue swinging according to the primary manipulation type. In some embodiments, the processor **1922** may show the virtual representation of the object fall to the floor based on a highest slider value in operation **2026**. The virtual representation of the object can realistically drop to the floor using standard physics for a falling motion of the virtual representation. When the virtual representation of the object reaches another object or a ground level, the virtual representation can react with the other object or the ground level. For instance, a force could be applied to the other object based on the virtual representation dropping to the floor. The virtual representation can break or crack by the processor **1922** dividing the virtual representation into two or more part virtual representations. For instance, a picture frame can reach the ground level and cause a crack through the picture frame creating two separate parts of the picture frame. The two parts of the picture frame may operate independently for a remaining time of the simulation.

[0269] The virtual representation and boundaries can be shaken independently. For example, a first picture frame can move independently from a second picture frame and a wall. The processor **1922** can create a set of parent coordinate transforms in an application, such as Unity. The parent transform can be an intermediate coordinate system between the world coordinate and children coordinates. The processor **1922** can set all submeshes from all regions that have a first face classification, such as "floor", to a first transform (also referred to as a parent transform) and all submeshes from all regions that have a second classification, such as "wall", to a second transform (also referred to as a child transform). The processor **1922** can set meshes that belong to a plane selected by a User as well as a plane with render cracks to a third transform. The processor **1922** can apply separate random shaking to each of the parent transforms. Parent meshes ensure that a position of the sub meshes with respect to the parent transform remain constant. However, the position of the parent transform with respect to the world is allowed to change. This allows all meshes assigned to a common classification to move together.

[0270] The shaking and falling of objects, such as picture frames, can be performed by implementing Rigid body mechanics using a physic engine, such as a Unity Physics Engine. The selected wall plane and each picture frame can be assigned rigid body functionalities, which allows the objects to interact with each other in forms such as collisions, forces, constraints, and accelerations. The physics engine can provide functions that allow rigid bodies to relate to each other in a form of constraints. The physic engine can then iteratively solve multi-body simulation to give a next position of the rigid bodies given external forces. The overall steps for this procedure can include (1) assigning rigid bodies to a picture frame and a wall mesh, and (2) adding a revolute hinge joint constraint between picture frame and wall mesh.

[0271] The processor **1922** can determine a damage amount **2812** for the scene **2702** in operation **2914**. The damage amount **2812** can be calculated as the simulation is in progress and/or after the simulation of seismic activity is completed. The damage amount **2812** can be calculated as a percentage, fraction, value, etc. The damage amount **2812** can be individually calculated for each AR object. The processor **1922** can utilize specific effects applied to an object and an intensity of the effect applied. For instance, an object that falls twice a distance as a similar object could be determined to have more damage. The processor **1922** can also determine a fragility of an object. For example, the processor **1922** can assign more damage to a vase falling an equal distance to a book.

[0272] Although FIG. **29** illustrates an example method for an AR earthquake visualization and assessment process **2900**, various changes may be made to FIG. **29**. For example, while shown as a series of steps, various steps in FIG. **29** may overlap, occur in parallel, occur in a different order, or occur any number of times.

[0273] FIGS. **30**A-**30**C illustrate one embodiment of an AR earthquake visualization and assessment system **3000** in accordance with this disclosure, namely, from the viewpoint of a user holding or directing a mobile device **3002** (the user's hand is not shown) scanning one or more physical scenes with device sensors **3018** and displaying one or more corresponding AR scenes generated by the system on the device screen **3016**. In this embodiment, the system **3000** is used to visualize the effects of a simulated earthquake on a first room **3008** (FIGS. **30**A and **30**C) and a second room **3012** (FIG. **30**B) separated by a boundary **3014**. For illustrative purposes, respective overhead schematic views (each denoted a “Plan Diagram”) are provided for each AR scene showing the relative positions of the first room **3008** (denoted “A”) and the second room **3012** (denoted “B”) and the user's viewpoint and direction of view (denoted with arrow “V”) and field of view (denoted with dotted lines) before and after the simulated earthquake.

[0274] The AR earthquake visualization and assessment system **3000** can display an AR earthquake scene **3004** (FIG. **30**C) based on a first scene **3006** captured in a first room **3008** (FIG. **30**A) visible behind the mobile device and a second scene **3010** captured in a nearby second room **3012** (FIG. **30**B) separated from the first room by a boundary **3014** (in this case, a wall). FIG. **30**A illustrates the user's point of view as the user directs a mobile device **3002** towards the first scene **3006** of the first room **3008** with a boundary **3014**. In this embodiment, the boundary **3014** is a wall, however, in other embodiments a boundary can be any surface defining the extent of a room, including walls, floors, ceilings, windows, doors, etc. The boundary **3014** can also be partial surfaces such as folding screens, curtains, sliding doors, divider screens, etc. that defines the visible extent of a room.

[0275] In FIG. **30**A, the first scene **3006** illustrates how the first room **3008**, the objects within, and the boundary **3014** actually look while the screen **3016** displays how the first AR scene **3007** generated by the system **3000** looks prior to a simulated earthquake or seismic event. FIG. **30**B illustrates the user's point of view as the user directs the mobile device **3002** towards the second scene **3010** in the second room **3012** on an opposite side of the boundary **3014** from the first room **3008**. In FIG. **30**B, the second scene **3010** illustrates how the second room **3012**, the objects within, and the boundary **3014** actually look while the screen **3016** displays how the second AR scene **3011** generated by the system **3000** looks prior to the simulated earthquake or seismic event. FIG. **30**C illustrates the user's point of view after the system **3000** has incorporated the effects of the simulated earthquake or seismic event into the earthquake visualization scene **3030** displayed on the screen **3016**, with the user directing the mobile device **3002** towards the first scene **3006** of the first room **3008**. In FIG. **30**C, the first scene **3006** is unchanged from that illustrated in FIG. **30**A because the first room **3008**, the contents within, and the boundary **3014** have not physically changed due to the simulated earthquake or seismic event. However, the generated AR scene **3030** now shown on device screen **3016** visualizes how the earthquake event that was simulated could affect the rooms **3008** and **3012**, their contents, and the boundary **3014**. For example, in the visualization **3030** displayed on the screen **3016**, the AR boundary **3031** appears broken to form an AR aperture **3054** that allows the user to visualize from the first AR scene **3007**, through the AR aperture, and into the second AR scene **3011**. Further, the screen **3016** can display the AR objects within the AR scenes **3007** and **3011**, e.g., AR chairs **3036**, AR pictures (or paintings) **3032**, **3034** and **3050**, AR beds **3046**, AR pillows **3048**, AR lamps **3052** as being displaced, upset, and/or damaged by the simulated earthquake event. In some embodiments, visualization **3030** can display the AR objects from one AR scene displaced through the AR apertures **3054** from one AR scene to another AR scene, or even visible across both AR scenes. In some embodiments, the AR rooms, AR objects or AR boundaries can be modified from an undamaged appearance to show simulated damage **3058** including, but not limited to, AR cracks or AR 3D cracks resulting from the simulated earthquake event.

[0276] As shown in FIGS. **30**A, the mobile device **3002** can include a display **3016**, one or more sensors **3018** (e.g., optical camera, time of flight camera, etc.), and one or more physical inputs **3020** (e.g., buttons, switches, etc.). The one or more sensors **3018** can have a scan area **3022** used to capture the first scene **3006** while scanning the first room **3008**. The one or more sensors **3018** can capture physical boundaries of the first room **3008**, including boundary **3014**, and physical objects within including, but not limited to, a first picture **3024**, a second picture **3026**, and a chair **3028**. The mobile device **3002** can create virtual or AR representations **3007** of the first scene **3006** prior to the simulated earthquake and present AR representations **3007** of the scene on the display **3016**. For example, the display **3016** in FIG. **30**A shows an AR boundary **3031**, an AR first picture **3032**, an AR second picture **3034**, and an AR chair **3036** corresponding to the physical boundary **3014**, the physical first picture **3024**, the physical second picture **3026**, and the physical chair **3028**.

[0277] As shown in FIG. **30**B, the one or more sensors **3018** can capture physical boundaries of the second room **3012**, including an opposite side of the boundary **3014**, and physical objects including, but not limited to, a bed **3038**, a pillow **3040**, a painting **3042**, and a lamp **3044**). The system **3000** running on the mobile device **3002** can create virtual or AR representations of the second scene **3010** and present the representations on the display **3016**. For example, the display **3016** in FIG. **30**B shows an opposite side of the AR boundary **3031**, an AR bed **3046**, an AR pillow **3048**, an AR painting **3050**, and an AR lamp **3052** corresponding to the respective physical boundary **3014**, the physical bed **3038**, the physical pillow **3040**, the physical painting **3042**, and the physical lamp **3044**.

[0278] The AR boundaries and AR objects generated by the system **3000** can have characteristics individually determined and applied during and/or after the scanning process. For example, a weight, material, roughness factor, etc. can be estimated for an AR object. The characteristics can be used for each manipulation of the AR boundary and the AR objects. In certain embodiments, a value can be assigned to each of the AR boundaries and the AR objects using the system **3000**. The value can be based on a purchase value, an actual value, an input value, a replacement value, etc. The values can be stored in a table, such as inventory table **2800** shown below in FIGS. **28**A and **28**B, or in the memory of the mobile device **3002**, which can be viewed separately or in combination with the AR environment. The values can also be shown on the screen **3016** with AR objects as each AR object is selected or as the value is determined. The values that are shown on the screen **3016** with the AR objects can disappear after a period of time or on approval by the user. The values can be adjusted by the user. The scenes **3006** and **3010** can include any other type of object(s) and any number of a specific object.

[0279] As shown in FIG. **30**C, the AR earthquake visualization and assessment **3000** can include scanning the first scene **3006** using the mobile device **3002**. The first scene **3006** includes the boundary **3014**, the first picture **3024**, the second picture **3026**, and the chair **3028** but does not include the bed **3038**, the pillow **3040**, the painting **3042**, and the lamp **3044**. The system **3000** of FIG. **30**C is shown generating an AR earthquake scene and showing it on the display screen **3016** based on an AR visualization of the damaged first room **3008** being visible in the AR foreground behind the mobile device **3002** and an AR visualization of the damaged second room **3012** being visible in the AR background through some damage (i.e., the aperture **3054**) caused to the AR boundary **3031**.

[0280] Values for damage can be determined by the system **3000** and used to determine which of the AR effects are applied to a respective AR boundary or AR object. The values for damage can be based on a repair value, a replacement value, etc. The values for damage can be calculated based on an estimated amount of overall damage. For an amount of damage exceeding a threshold, an AR object can be considered fully damaged or destroyed. The values for damage can be saved in the table in the memory of the mobile device **3002**. The damage values can be shown on the respective AR boundary or AR object that receives the damage. The damage values can disappear after a period of time or selection by a user. The damage values can include a total damage for the seismic activity applied to AR representation of the scenes **3006** and **3010**.

[0281] The AR boundary **3031** and AR objects **3032-3036** and **3046-3052** can be manipulated on the display **3016** while remaining unaffected in the first scene **3006**. As the mobile device **3002** is moved around the room, the display **3016** updates a current status of the AR objects **3032-3036** and **3046-3052**. The one or more sensors **3018** can capture the actual room **3008**, **3012** and the display **3016** can show the corresponding AR scenes in a current state of simulated seismic activity. The display **3016** can also display controls (such as a seismic attribute control **2740** and a start button **2742** shown in FIG. **27**).

[0282] The display can apply an aperture effect **3054** to the AR boundary **3031**. The aperture effect **3054** can be applied to one or more boundaries between rooms that have been scanned. The aperture effect **3054** allows for viewing the AR objects in the second room **3012** even though the second room **3012** is not visible though the actual boundary **3014** between the first room **3008** and the second room **3012** is fully intact and obscures the view to the second room **3012**. Even when the actual boundary **3014** is partially transparent, the area of the aperture effect **3054** does not obscure the AR objects and can have a different transparency level from the AR boundary **3031**. For example, the AR bed **3046** and the AR pillow **3048** can be seen in the second scene **3010** through the aperture effect **3054**, while the actual bed **3038** and pillow **3040** cannot be seen in the second scene **3010** through the boundary **3014**.

[0283] The aperture effect **3054** also allows for AR objects **3032-3036** in the first room **3008** to be partially or fully displaced into the second room **3012** and for AR objects **3046-3052** in the second room **3012** to be partially or fully displaced into the first room **3008**. For example, the AR lamp **3052** is illustrated lying on the ground across an AR border **3056** between the first scene **3006** and the second scene **3010**. The aperture effect **3054** can place a mask in the display over the actual boundary **3014** and display generated AR boundaries and AR objects from the second scene **3010** over the mask in the correct scale and orientation to appear that the second scene **3010** is view through the aperture in the common boundary.

[0284] The display **3016** can apply a vibration or shaking effect to an AR boundary **3031** and AR objects **3032-3036** and **3046-3052**. The shaking effect can be applied equally to the AR boundary **3031** or can be applied stronger or weaker to other AR boundaries. The shaking effect applied to the AR boundary **3031** can be additionally applied to each of the AR objects **3032-3036** and **3046-3052** in a manner of equal characteristics or reduced effectiveness characteristics. The reduced effectiveness characteristics can include lower shaking intensity transferred from the AR boundary **3031**, lower resistance (increased duration of shaking), etc. An amount of the vibration for the AR boundary **3031** or AR objects **3032-3036** and **3046-3052** can be controlled based on the input **2744** to the seismic attribute control **2740**.

[0285] The display **3016** can apply a cracking effect **3058** on an AR boundary, such as the AR boundary **3031**, **3060**. The cracking effect **3058** can cause a 3D crack to form in the AR boundary. The 3D crack can be viewed from different angles to show a depth to the 3D crack. The cracking effect **3058** can include a 3D crack that extends from a first AR boundary to a second AR boundary. For example, the cracking effect **3058** can begin on an AR wall and extend to an AR ceiling, an AR second wall, and/or the AR floor. A 3D crack on an AR boundary, such as the AR wall, can extend behind and be hidden by an AR object to continue extending past the AR object in view on the display **3016**. A size and growth rate for the 3D crack on the AR boundary or AR object can be controlled based on the input **2744** to the seismic attribute control **2740**. A 3D crack can extend from an edge of the aperture effect **3054**. The cracking effect **3058** can be viewed on AR boundaries **3061** in a second scene **3010** in the display **3016** through the AR aperture effect **3054** although the corresponding physical boundaries **3060** are not visible by the mobile device **3002** from the first room **3008**.

[0286] The display **3016** can apply a swinging effect to AR objects **3032-3036** and **3046-3052**, such as the AR first picture **3032**. The swinging effect can cause the AR object to shake similarly to the AR boundary **3031** contacting the AR object due to a fixed pendulum that the swinging occurs about. The swinging effect on the AR object can include a translation component equal to the vibration or shaking effect of the AR boundary. A swing rate for the swinging of AR object can be controlled based on the input **2744** to the seismic attribute control **2740**.

[0287] The display **3016** can apply a falling effect to AR objects **3032-3036** and **3046-3052**, such as the AR second picture **3034**. The falling effect can move the AR object separately from the AR boundary and other AR objects. The falling effect can be based on gravity for controlling a falling rate and/or a falling speed. The falling effect can use an estimated or determined weight of the AR object to determine a force that could be applied to other AR objects and/or AR boundaries. A timing for the falling of AR object can be controlled based on the input **2744** to the seismic attribute control **2740**.

[0288] In certain embodiments, a secondary effect can be applied to AR objects **3032-3036** and **3046-3052**, such as the AR second picture **3034**. For example, a falling effect on an AR object causes the AR object to forcefully contact an AR boundary, such as the AR floor, or another AR object, such as the AR chair **3036**. A secondary effect of the falling effect could be a breaking effect. The display can apply the breaking effect to the AR object, such as the AR second picture **3034**. Other examples of secondary effects can include, a bouncing effect, a ricochet effect, a merge effect, striking effect, etc.

[0289] The breaking effect can cause an AR boundary **3031** and AR objects **3032-3036** and **3046-3052** to split into multiple AR partial objects. Each of the multiple AR partial objects can be smaller than the AR object before the split. In certain embodiments, the multiple AR partial objects can collectively be approximately equal to the broken AR object. An amount of AR partial objects can be based on an original size of an AR object, a force applied to the AR object during simulation from one or more other AR effects, etc. AR partial objects from the breaking effect can be displayed on both sides of the AR border **3056**. In other words, the AR partial objects from a single AR object **3032-3036** and **3046-3052** can be displayed in the first scene **3006** and the second scene **3010**.

[0290] A shatter effect can cause an AR object to have multiple AR partial objects split from a main AR object, such as an AR window or an AR mirror. The shatter effect can be a primary effect from the seismic damage or a secondary effect from another AR object forcefully interacting with the original AR object. For example, the AR window can shatter due directly to the vibrations from a seismic event or from an AR object, such as the AR second picture **3034** or generation of a random AR object, such as an AR branch of a tree, an AR power line, etc., falling through the AR object, such as AR window. AR partial objects from the shatter effect can be displayed on both sides of the AR border **3056**. In other words, the AR partial objects from a single AR object **3032-3036** and **3046-3052** can be displayed in the first scene **3006** and the second scene **3010**.

[0291] Although FIG. **30** illustrates an example AR earthquake visualization and assessment **3000**, various changes may be made to FIG. **30**. For example, various components in FIG. **30** may be combined, further subdivided, replicated, omitted, or rearranged and additional components may be added according to particular needs.

[0292] FIG. **31** illustrates an example method for an AR earthquake visualization and assessment process **3100** in accordance with this disclosure. As shown in FIG. **31**, the processor **1922** can scan multiple scenes **3006** and **3010** using an optical recognition method in operation **3102**. In certain embodiments, the first and second scenes **3006** and **3010** can be interiors of respective rooms and one or more boundaries **3014**, **3060** of the room can be scanned. The mobile device **3002** can further include a time-of-flight camera or Lidar sensor **1916**, **3018**, which the processor **1922** can operate to scan the one or more boundaries of each room. The processor **1922** can determine a distance to different points on a wall, ceiling and/or floor using Lidar. In some embodiments, the operation **3102** includes detecting planes in the scenes **3006** and **3010** using the depth information from the optical camera and/or Lidar camera **1916**. In some embodiments, the detecting planes in the scenes **3006** and **3010** of the operation **3102** further includes determining, using the depth information from the scan, that planes detected in the scenes **3006** and **3010** are one or more walls, ceilings and/or floors present in the scenes **3006** and **3010**. In some embodiments, the determining that planes detected in the scenes **3006** and **3010** are one or more walls, ceilings and/or floors present in the scenes **3006** and **3010** further comprises using 3-D camera position information or 6-D camera position and orientation information in association with the depth information.

[0293] The processor **1922** can also operate the camera **1916**, **3018** to capture the boundaries, such as wall, ceiling, or floor, as an orientation of the mobile device **3002** while being rotated and moved around the room. One or more sensors of the mobile device **3002** can capture details for determining a camera pose for the camera **1916** as each frame of the boundaries, such as wall, ceiling, or floor, are captured. The camera pose can be saved along with the Lidar information or each could be stored separately with a respective time stamp to match.

[0294] The processor **1922** can identify boundaries **3014**, **3060** and objects in scenes **3006** and **3010** in operation **3104**. The processor **1922** can scan the scenes **3006** and **3010** for meshes. The meshes can be defined and/or categorized by the processor **1922**. The meshes can also be associated with a specified plane or specified planes by the processor **1922**. Meshes can be associated with objects in the scene in addition to a boundary. For example, meshes can be defined as part of a wall and part of one or more picture frames on the wall. In order to identify boundaries, the processor **1922** can assign one or more meshes from a region to the plane. The processor **1922** can use the extracted camera image to determine a plane for each of the meshes. The meshes can be classified and associated with a specified plane by the processor **1922**. The processor **1922** can also identify boundaries that are between or are common to different scenes, such as boundary **3014**, **3060**. The processor **1922** can identify a boundary **3014** that divides a first room **3008** and a second room **3012**.

[0295] The processor **1922** can extract object suggestions from the mesh. The processor **1922** can identify meshes corresponding to one or more objects in the scenes **3006** and **3010**. The meshes of different objects in the scene can overlap and the processor **1922** can distinguish different objects. The processor **1922** can extract objects from the mesh belonging to a selected scene. Meshes that are not assigned to one or more of the planes, backgrounds, walls, etc. can be determined by the processor to be objects. The objects can be individually processed by the processor **1922** for details of the objects, such as dimensions, orientation, position, etc.

[0296] The processor **1922** can count the objects in operation **3106**. Counting the objects can include taking an inventory of the objects in the scenes **3006** and **3010** and placing the information in an inventory table **2800**. The objects can be assigned an object number **2802** that can be used to relate a row on the table to an AR object by the processor **1922**. A description **2804** of the objects can be determined while counting. The description **2804** of the objects can allow a user to quickly and easily determine which row corresponds to a specific AR object or object in a scene. The description **2804** can include a name of the object, a position of the object, characteristics of the object, etc. The counting of the objects can also include an accounting of the objects and an estimated value **2806** or other value can be determined.

[0297] The processor **1922** can create one or more AR boundaries **3031**, **3061** and one or more AR objects from the boundary and objects identified in the scenes **3006** and **3010** in operation **3108**. The processor **1922** can extract a color of a selected plane. In certain embodiments, the mobile device **3002** can receive an input from a user for selecting a color of the selected plane, such as a tap on the display **1920** of the mobile device **3002**. The camera can provide a camera image and the processor **1922** can determine a location in the camera image corresponding to a location of the tap on the display **1920**. The processor **1922** can determine a color at the location in the camera image to extract. In certain embodiments, the processor **1922** can extract a color based on other factors, such as amount of the color in the scene, an average color of the wall, etc. The processor **1922** can set a color for a mesh according to the extracted color. The processor **1922** can take the extracted color and apply the color to the mesh corresponding to the boundary of the scene. In certain embodiments, the processor **1922** can apply the extracted color to a wall, a floor, and/or a ceiling in a room.

[0298] The processor **1922** can extract images for an object in a camera frame. The processor **1922** can extract one or more camera images supplied by the AR camera. The camera images can be continuously captured based on movement of the AR camera in order to capture the object from different angles. In certain embodiments, the processor **1922** can capture camera images including one or more picture frames. The processor **1922** can determine that vertexes of the objects are clearly visible in the camera frame. Using the AR camera, the processor **1922** can identify each vertex of an object shown in the camera image. Camera images where the object is only partially in the camera image can be skipped or used for clearly identified vertexes. The processor **1922** can extract an image corresponding to the object as in operation **2128**. The details of the objects can be used by the processor **1922** to identify an actual object in the camera image and extract a partial image for the actual object. For example, the processor **1922** can identify a picture frame in one or more camera images and make sure that all the corners of the picture frame are clearly visible in each camera image. The processor **1922** can instantiate selected objects as in operation **2130**. The processor **1922** can overlay the actual image of the object over the corresponding meshes associated with the object in the AR scene. For example, an actual image of a picture frame can be overlaid over meshes associated with the picture frame.

[0299] The processor **1922** can receive seismic characteristics in operation **3110**. The seismic characteristics can be received from a user, read from a memory of the mobile device **3002**, received from another mobile device **3002**, or received from a remote server. For example, seismic characteristics or a seismic algorithm can be stored in a memory of the mobile device **3002**. The user can be presented an input, such as a slider, to control a seismic characteristic, such as a seismic intensity. The slider values can be controls for a parameter or attribute of the seismic activity. For example, the slider can control an intensity, a time period, etc. Additional parameters or attributes of the seismic activity not controlled by a slider can have values entered prior to the simulation or read from a memory **1918**. The processor **1922** can show a slider or other mechanism for selecting the parameter value or attribute of the seismic activity on the display **1920**. In certain embodiments, the slider can control a current intensity and seismic activity would continuously occur at the selected current intensity until the current intensity is set to zero on the slider. In some embodiments, a touch-screen or physical buttons, voice commands or other inputs can be used for entering parameters or attributes of the seismic activity instead of, or in addition to, the slider.

[0300] The processor **1922** can display seismic effects based on the seismic characteristics in operation **3112**. The processor **1922** can generate simulated (e.g., AR) seismic activity for the scene. The AR seismic activity can be shown to the user through the display **1920** or **3016** of the mobile device **3002**. The AR seismic activity can cause the portion of the AR boundary that is currently visible to shift in the frame of view on the display **1920** or **3016**. The edge of the boundary can be moved in and out of the frame in the display according to the one or more slider values. In certain embodiments, the AR seismic activity can also be applied to AR objects in other rooms corresponding to actual objects that are not visible to the mobile device **3002**. In certain embodiments, the seismic effects can be viewed in each room after the seismic activity is concluded in the simulation. The seismic activity can also be restarted or continuous as the mobile device **3002** is moved from room to room.

[0301] One example of a seismic effect is an aperture effect **3054** to the AR boundary **3031**. The aperture effect **3054** can be applied to one or more boundaries between rooms that have been scanned. The aperture effect **3054** allows for viewing the AR objects in the second room **3012** even though second room **3012** is not visible though the actual boundary **3014** between the first room **3008** and the second room **3012** is fully intact and obscures the view to the second room **3012**. Even when the actual boundary **3014** is partially transparent, the area of the aperture effect **3054** does not obscure the AR objects and can have a different transparency level from the AR boundary **3031**. For example, the AR bed **3046** and the AR pillow **3048** can be seen in the second scene **3010** through the aperture effect **3054**, while the actual bed **3038** and pillow **3040** cannot be seen in the second scene **3010** through the boundary **3014**.

[0302] The processor **1922** can display simulated (i.e., AR) cracks in the AR boundary shown on the display **3016** based on a slider value. In some embodiments, the processor **1922** can dynamically generate AR cracks in the AR boundaries during a simulated seismic event, i.e., the cracks may grow in length or width as the seismic event progresses. In some embodiments, the processor **1922** can create a random number of AR cracks with varying paths, widths and/or depths based on the slider value. In some embodiments, the processor **1922** can generate AR cracks featuring parallax shading to give the cracks the appearance of three-dimensional depth and width based on the point of view. The selected intensity of the of the simulation can be a factor for the crack generation. The crack generation can also be determined based on an amount of time that has passed and/or a total time of the simulation. For example, additional cracks may be formed as the time of the simulated seismic activity continues. In addition, the dimension of the cracks can change over time and based on the factors of the seismic activity.

[0303] The processor **1922** can move or manipulate an AR object (i.e., the virtual representation of the object) shown on the display **1920**, **3016** during a simulation of a seismic event. In some embodiments, the processor **1922** may determine the type of a movement or a manipulation to be applied to an AR object based on one or more seismic characteristics input by the user or received from the system. In some embodiments, the processor **1922** may determine the type of a movement or a manipulation to be applied to an AR object based on the category of like objects into which the object has been classified. In some embodiments, the processor **1922** may determine the type of a movement or a manipulation to be applied to an AR object based on specified or sensed characteristics of the AR object including, but not limited to the object's mass, size and/or material. In some embodiments, the processor **1922** may determine the type of a movement or a manipulation to be applied to the AR object by applying a combination of the factors described above. Types of movements of the AR object that may be applied by the processor **1922** and shown on the display **1920**, **3016** include, but are not limited to, translation, rotation, oscillation, falling and rolling. Type of manipulations of the AR object that may be applied by the processor **1922** and shown on the display **1920**, **3016** include, but are not limited to, cracking, breaking, burning, leaking, sparking and soiling. For example, if an object in the scene is classified as a type of object that may be manipulated with "oscillation," the processor **1922** can show the AR object swinging back and forth about a hinge point during the seismic event, and the magnitude of the swinging may be selected based on a seismic characteristic set by the slider value and/or on a characteristic of the object, such as its size. The virtual representation of the object can swing in relation to the boundaries of the scene. In addition, if there are multiple AR objects in a scene, each virtual representation may swing independently from other virtual representations. In certain embodiments, a virtual representation of a picture frame can swing on a wall. While the virtual representation is swinging, the actual object is obscured by the wall set as a boundary in order for the swinging motion to not be disrupted by the actual picture frame.

[0304] The processor **1922** can apply secondary manipulation types to the AR objects during a simulation of a seismic event. A secondary manipulation can be any manipulation that is applied only when predetermined conditions are met during a simulated seismic event. For example, if the AR object has a primary manipulation type of “oscillation” during a seismic event, the AR object may have a secondary manipulation type of “fall” if the magnitude of the shaking exceeds a preselected level, or if the shaking duration exceeds a preselected period. Thus, the processor **1922** will initially show the AR object oscillating (i.e., swinging) during a simulated seismic event, but will only show the AR object falling if the preselected conditions are met. If the preselected conditions are not met, the AR object will continue swinging according to the primary manipulation type. In some embodiments, the processor **1922** may show the virtual representation of the object fall to the floor based on a highest slider value as in operation **2026**. The virtual representation of the object can realistically drop to the floor using standard physics for a falling motion of the virtual representation. When the virtual representation of the object reaches another object or a ground level, the virtual representation can react with the other object or the ground level. For instance, a force could be applied to the other object based on the virtual representation dropping to the floor. The virtual representation can break or crack by the processor **1922** dividing the virtual representation into two or more partial virtual representations. For instance, a picture frame can reach the ground level and cause a crack through the picture frame creating two separate parts of the picture frame. The two parts of the picture frame may operate independently for a remaining time of the simulation.

[0305] The virtual representation and boundary can be shaken independently. For example, a first picture frame can move independently from a second picture frame and a wall. The processor **1922** can create a set of parent coordinate transforms in an application, such as Unity. The parent transform can be an intermediate coordinate system between the world coordinate and children coordinates. The processor **1922** can set all submeshes from all regions that have a first face classification, such as “floor”, to a first transform (also referred to as a parent transform) and all submeshes from all regions that have a second classification, such as “wall”, to a second transform (also referred to as a child transform). The processor **1922** can set meshes that belong to a plane selected by a User as well as a plane with rendered cracks to a third transform. The processor **1922** can apply separate random shaking to each of the parent transforms. Parent meshes ensure that a position of the sub meshes with respect to the parent transform remain constant. However, the position of the parent transform with respect to the world is allowed to change. This allows all meshes assigned to a common classification to move together.

[0306] The shaking and falling of objects, such as picture frames, can be performed by implementing Rigid body mechanics using a physics engine, such as a Unity Physics Engine. The selected wall plane and each picture frame can be assigned rigid body functionalities, which allows the objects to interact with each other in forms such as collisions, forces, constraints, and accelerations. The physics engine can provide functions that allow rigid bodies to relate to each other in a form of constraints. The physics engine can then iteratively solve a multi-body simulation to give a next position of the rigid bodies given external forces. The overall steps for this procedure can include (1) assigning rigid bodies to a picture frame and a wall mesh, and (2) adding a revolute hinge joint constraint between picture frame and wall mesh.

[0307] The processor **1922** can determine a damage amount **2812** for the scenes **3006** and **3010** in operation **3114**. The damage amount **2812** can be calculated as the simulation is in progress and/or after the simulation of seismic activity is completed. The damage amount **2812** can be calculated as a percentage, fraction, value, etc. The damage amount **2812** can be individually calculated for each AR object. The processor **1922** can utilize specific effects applied to an object and an intensity of the effect applied. For instance, an object that falls twice a distance as a similar object could be determined to have more damage. The processor **1922** can also determine a fragility of an object. For example, the processor **1922** can assign more damage to a vase falling an equal distance to a book.

[0308] Although FIG. **31** illustrates an example method for an AR earthquake visualization and assessment process **3100**, various changes may be made to FIG. **31**. For example, while shown as a series of steps, various steps in FIG. **31** may overlap, occur in parallel, occur in a different order, or occur any number of times.

[0309] It will be appreciated by those skilled in the art having the benefit of this disclosure that these methods and systems for display of an electronic representation of physical effects and property damage resulting from a parametric natural disaster event provides an augmented reality presentation of the personalized effect of an earthquake on actual objects around a user. It should be understood that the drawings and detailed description herein are to be regarded in an illustrative rather than a restrictive manner, and are not intended to be limiting to the particular forms and examples disclosed. On the contrary, included are any further modifications, changes, rearrangements, substitutions, alternatives, design choices, and embodiments apparent to those of ordinary skill in the art, without departing from the spirit and scope hereof, as defined by the following claims. Thus, it is intended that the following claims be interpreted to embrace all such further modifications, changes, rearrangements, substitutions, alternatives, design choices, and embodiments.

Claims

1. A method for displaying an augmented reality (AR) representation of physical effects and property damage resulting from a parametric earthquake event, comprising: scanning, using one or more sensors of a user device, first and second scenes with a common boundary; identifying the common boundary and objects in the first and second scenes; creating an AR boundary for the common boundary of the first and second scenes and AR objects for the objects in the first and second scenes; receiving at least one seismic characteristic from a user of the user device; and while scanning the first scene with the user device, displaying an aperture effect on the AR boundary and at least one seismic effect on the AR objects in the second scene based on the at least one received seismic characteristic, wherein the aperture effect allows viewing the at least one seismic effect on AR objects in the second scene on the user device while the second scene is not visible to the one or more sensors of the user device.

2. A method in accordance with claim 1, wherein the at least one seismic effect causes AR objects corresponding to objects in the first scene to pass through the aperture effect in the common boundary into the second scene.

3. A method in accordance with claim 1, wherein the at least one seismic effect causes AR objects corresponding to objects in the second scene to pass through the aperture effect in the common boundary into the first scene, where the objects in the second scene are not visible to the user device.

4. A method in accordance with claim 1, wherein the at least one seismic effect causes AR objects corresponding to objects in the second scene to partially pass through the aperture effect in the common boundary into the first scene, where the objects in the second scene are not visible to the user device.

5. A method in accordance with claim 1, wherein displaying the at least one seismic effect on the AR objects comprises: displaying the at least one seismic effect on AR objects in the second scene through aperture in the common boundary.

6. A method in accordance with claim 1, wherein the aperture effect in the common boundary partially covers the common boundary.

7. A method in accordance with claim 1, wherein the at least one seismic effect is a cracking effect that extends from an edge of the aperture effect on the common boundary.

8. A non-transitory computer readable medium, for displaying an augmented reality (AR) representation of physical effects and property damage resulting from a parametric earthquake event, containing instructions that when executed cause a processor to: scan, using one or more sensors of a user device, first and second scenes with a common boundary; identify the common boundary and objects in the first and second scenes; create an AR boundary for the common boundary of the first and second scenes and AR objects for the objects in the first and second scenes; receive at least one seismic characteristic from a user of the user device; and while scanning the first scene with the user device, display an aperture effect on the AR boundary and at least one seismic effect on the AR objects in the second scene based on the at least one received seismic characteristic, wherein the aperture effect allows viewing the at least one seismic effect on AR objects in the second scene on the user device while the second scene is not visible to the one or more sensors of the user device.

9. A non-transitory computer readable medium in accordance with claim 8, wherein the at least one seismic effect causes AR objects corresponding to objects in the first scene to pass through the aperture effect in the common boundary into the second scene.

10. A non-transitory computer readable medium in accordance with claim 8, wherein the at least one seismic effect causes AR objects corresponding to objects in the second scene to pass through the aperture effect in the common boundary into the first scene, where the objects in the second scene are not visible to the user device.

11. A non-transitory computer readable medium in accordance with claim 8, wherein the at least one seismic effect causes AR objects corresponding to objects in the second scene to partially pass through the aperture effect in the common boundary into the first scene, where the objects in the second scene are not visible to the user device.

12. A non-transitory computer readable medium in accordance with claim 8, wherein displaying the at least one seismic effect on the AR objects comprises: displaying the at least one seismic effect on AR objects in the second scene through aperture in the common boundary.

13. A non-transitory computer readable medium in accordance with claim 8, wherein the aperture effect in the common boundary partially covers the common boundary.

14. A non-transitory computer readable medium in accordance with claim 8, wherein the at least one seismic effect is a cracking effect that extends from an edge of the aperture effect on the common boundary.

15. A user device comprising: one or more sensor configured to scan first and second scenes with a common boundary; and a processor configured to: identify the common boundary and objects in the first and second scenes; create an AR boundary for the common boundary of the first and second scenes and AR objects for the objects in the first and second scenes; receive at least one seismic characteristic from a user of the user device; and while scanning the first scene with the user device, display an aperture effect on the AR boundary and at least one seismic effect on the AR objects in the second scene based on the at least one received seismic characteristic, wherein the aperture effect allows viewing the at least one seismic effect on AR objects in the second scene on the user device while the second scene is not visible to the one or more sensors of the user device.

16. A user device in accordance with claim 8, wherein the at least one seismic effect causes AR objects corresponding to objects in the first scene to pass through the aperture effect in the common boundary into the second scene.

17. A user device in accordance with claim 8, wherein the at least one seismic effect causes AR objects corresponding to objects in the second scene to pass through the aperture effect in the common boundary into the first scene, where the objects in the second scene are not visible to the user device.

18. A user device in accordance with claim 8, wherein the at least one seismic effect causes AR objects corresponding to objects in the second scene to partially pass through the aperture effect in the common boundary into the first scene, where the objects in the second scene are not visible to the user device.

19. A user device in accordance with claim 8, wherein displaying the at least one seismic effect on the AR objects comprises: displaying the at least one seismic effect on AR objects in the second scene through aperture in the common boundary.

20. A user device in accordance with claim 8, wherein the aperture effect in the common boundary partially covers the common boundary.